LogLookup AI

Implementation, Lab Deployment and End-to-End Validation Report

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Project	LogLookup AI
Lab platform	UTM virtual machines on Apple Silicon
SIEM stack	Elasticsearch, Kibana, Fleet Server and Elastic Security
Endpoint telemetry	Windows 11 ARM, Elastic Agent and Sysmon

Editorial Note

This report is a refined version of the original project documentation. AI-assisted editing was used exclusively for language refinement, formatting, and presentation. All technical content, implementation, configuration, validation, screenshots, and findings are based on the author's original work and practical project execution.

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1. Executive summary

This report records the complete practical work carried out to build and validate the LogLookup AI lab. The work began with a three-machine virtual environment and ended with a working investigation pipeline that collected Windows telemetry, generated Elastic Security alerts, organised those alerts into an attack chain, applied risk scoring and AI-assisted reasoning, and presented the result in a dedicated investigation interface.

The lab used an Ubuntu virtual machine as the SIEM server, a Windows 11 ARM virtual machine as the monitored endpoint, and a Kali Linux virtual machine for controlled security testing. Elasticsearch, Kibana and Fleet Server were installed on Ubuntu. The Windows endpoint was enrolled through Elastic Agent and supplied Windows event logs and Sysmon telemetry. Detection rules were enabled in Elastic Security and validated against authentication, PowerShell, process, registry, network, scheduled-task and service activity.

After the Elastic environment had been validated manually, LogLookup AI was installed beside the SIEM. The application connected to Elasticsearch through an API key, selected the Elastic Security alert index, loaded its MITRE ATT&CK knowledge base and configured an AI provider. The final validation demonstrated the full path from endpoint activity to Elastic alert, correlated chain, AI verdict, evidence display and analyst-facing report.

2. Project scope and objectives

The purpose of the project was not simply to display SIEM alerts. The aim was to build a complete investigation workflow that could demonstrate how individual security detections can be turned into a coherent incident. This required both the underlying SOC laboratory and the LogLookup AI application to work together.

The practical objectives were to prepare the virtual machines, install the Elastic Stack, enrol endpoint agents, collect high-value Windows telemetry, enable suitable detection rules, generate controlled attack-like activity, inspect the raw events in Discover, review alerts in Elastic Security, create a timeline and case, then connect LogLookup AI and confirm that the same alert set could be processed into a structured attack chain.

3. Lab design

The laboratory was built in UTM on an Apple Silicon Mac. Ubuntu Server provided Elasticsearch, Kibana and Fleet Server. Windows 11 ARM acted as the monitored endpoint and generated the event stream used during the investigation. Kali Linux was prepared as a separate testing system for controlled activity where required. Keeping the roles on separate virtual machines made the lab closer to a small SOC environment than a single-host demonstration.

3.1 Pre-installation readiness check

The first step was to confirm that the machines, network connectivity and core services were ready. A pre-installation check prevents later troubleshooting from being confused by basic issues such as insufficient disk space, missing packages, incorrect host connectivity or unavailable services.

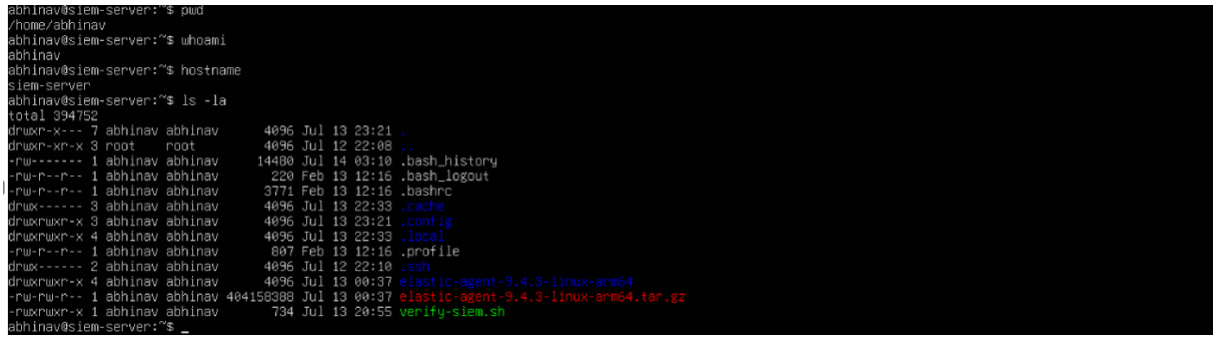


Figure 1. Initial readiness and environment checks completed before the SIEM and endpoint workflow was validated.

4. SIEM platform deployment

The SIEM server was built on Ubuntu and hosted Elasticsearch, Kibana and Fleet Server. Elasticsearch stored the telemetry and security alerts. Kibana provided the management and analyst interface. Fleet Server managed the Elastic Agents that forwarded data from the endpoint. This separation of responsibilities is important because it allows collection, storage, detection and investigation to be tested independently.

4.1 Detection content installed

Elastic Security detection rules were installed before generating activity. The rules provide the analytical layer that converts raw endpoint events into security alerts.

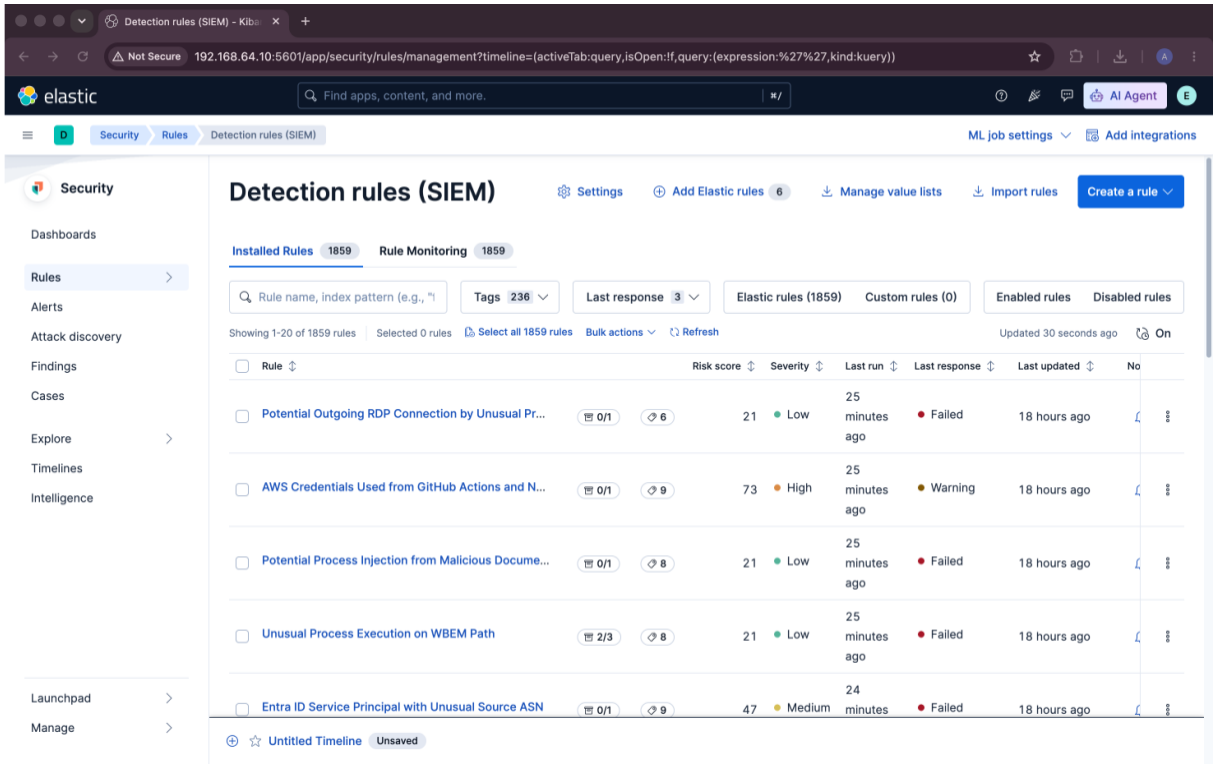


Figure 2. Elastic Security detection rules installed and available for validation.

4.2 Detection rules enabled

Installing rules is not enough; the required rules must be enabled and allowed to run. This step confirmed that Elastic could actively evaluate incoming telemetry.

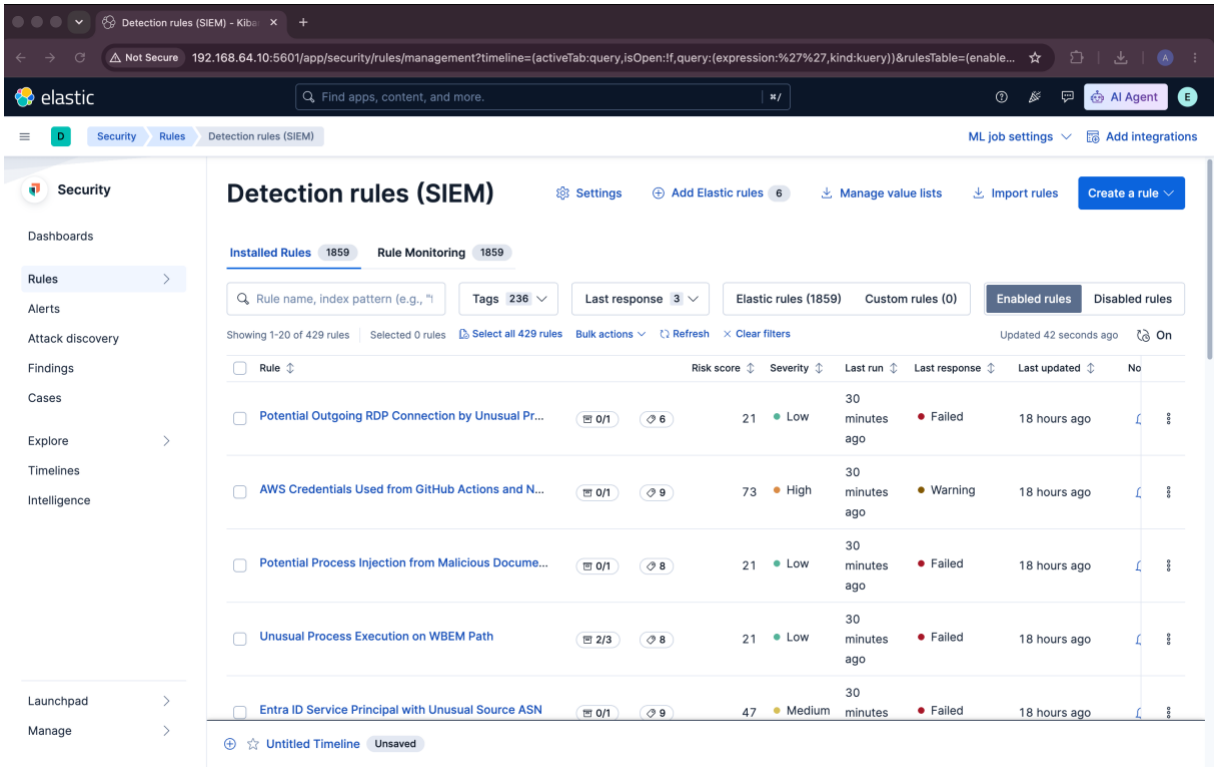


Figure 3. Detection rules enabled in Elastic Security.

4.3 Windows rule coverage

Windows-specific coverage was enabled because the victim machine generated the main endpoint data. The selected rules covered common execution, persistence and credential-related behaviour.

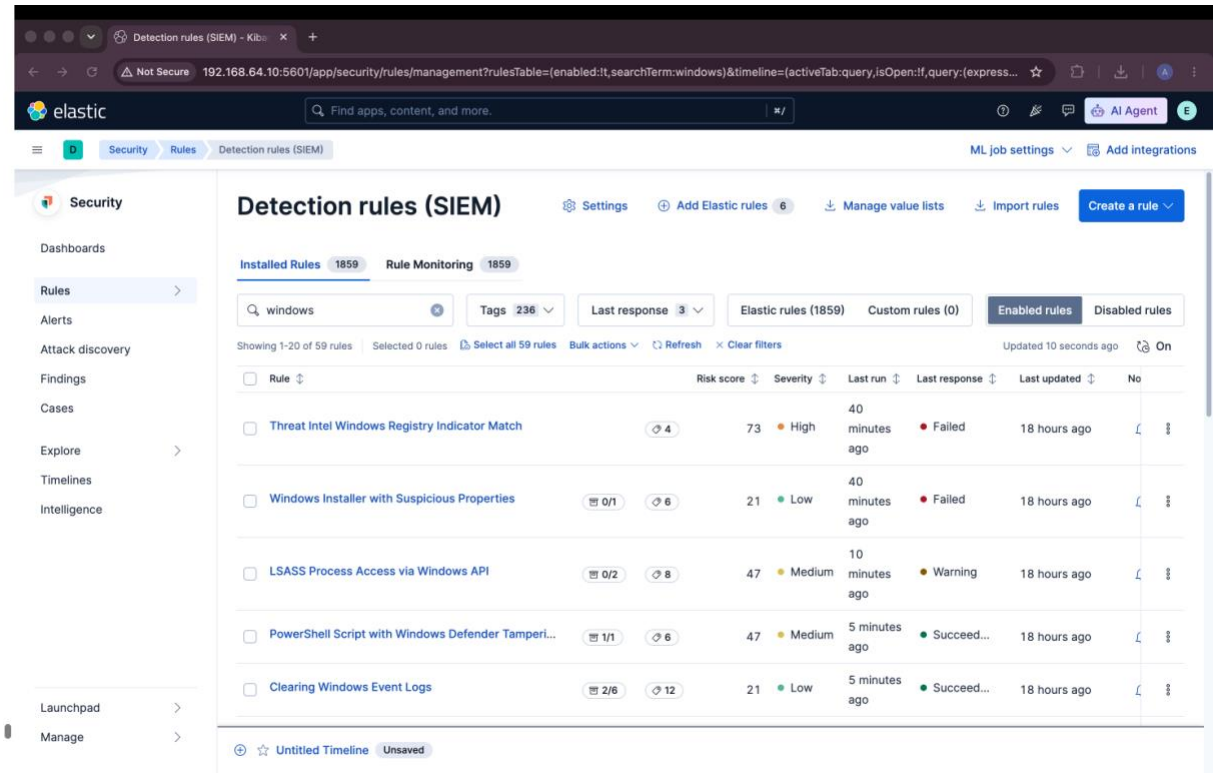


Figure 4. Windows-focused detection rules selected for the endpoint lab.

4.4 PowerShell rule coverage

PowerShell is a major source of legitimate administration and attacker activity. Enabling several PowerShell rules allowed the lab to validate encoded commands, suspicious arguments and script-based behaviour.

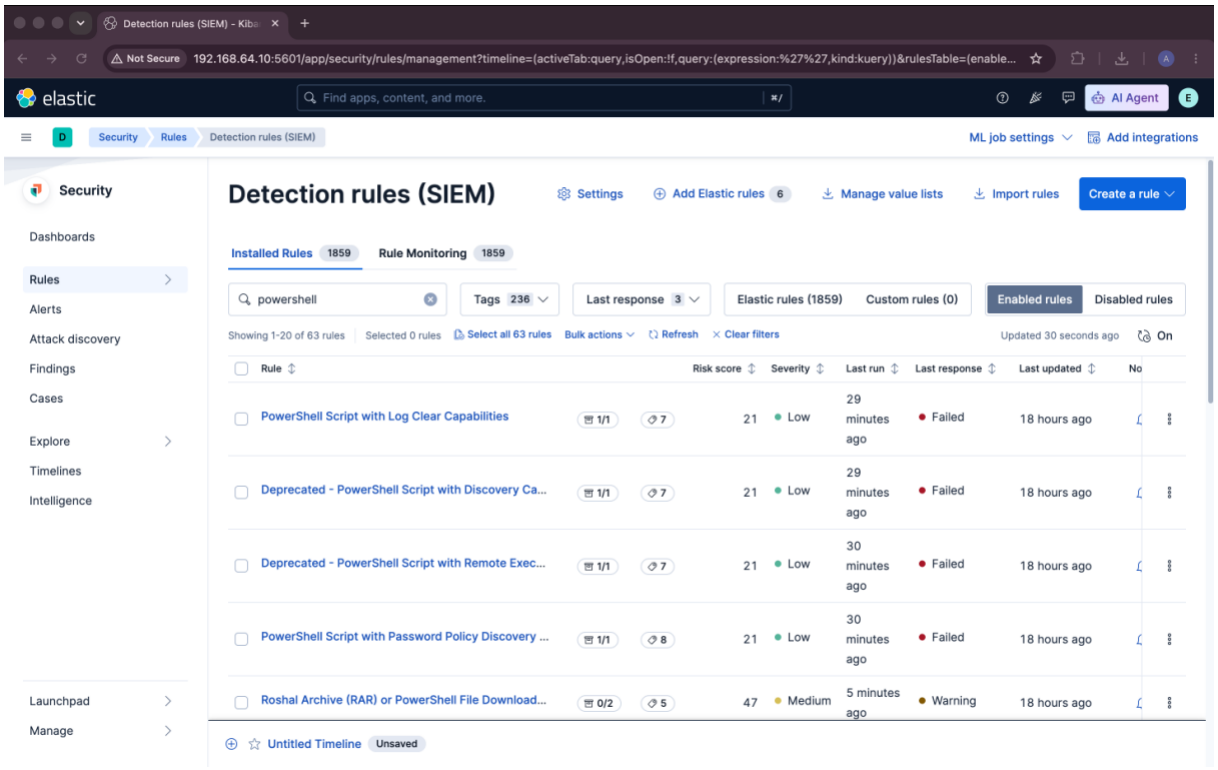


Figure 5. PowerShell detection rules enabled for script and command-line monitoring.

4.5 Credential rule coverage

Credential-focused rules were enabled to observe activity associated with token manipulation, suspicious credential access and related post-exploitation techniques.

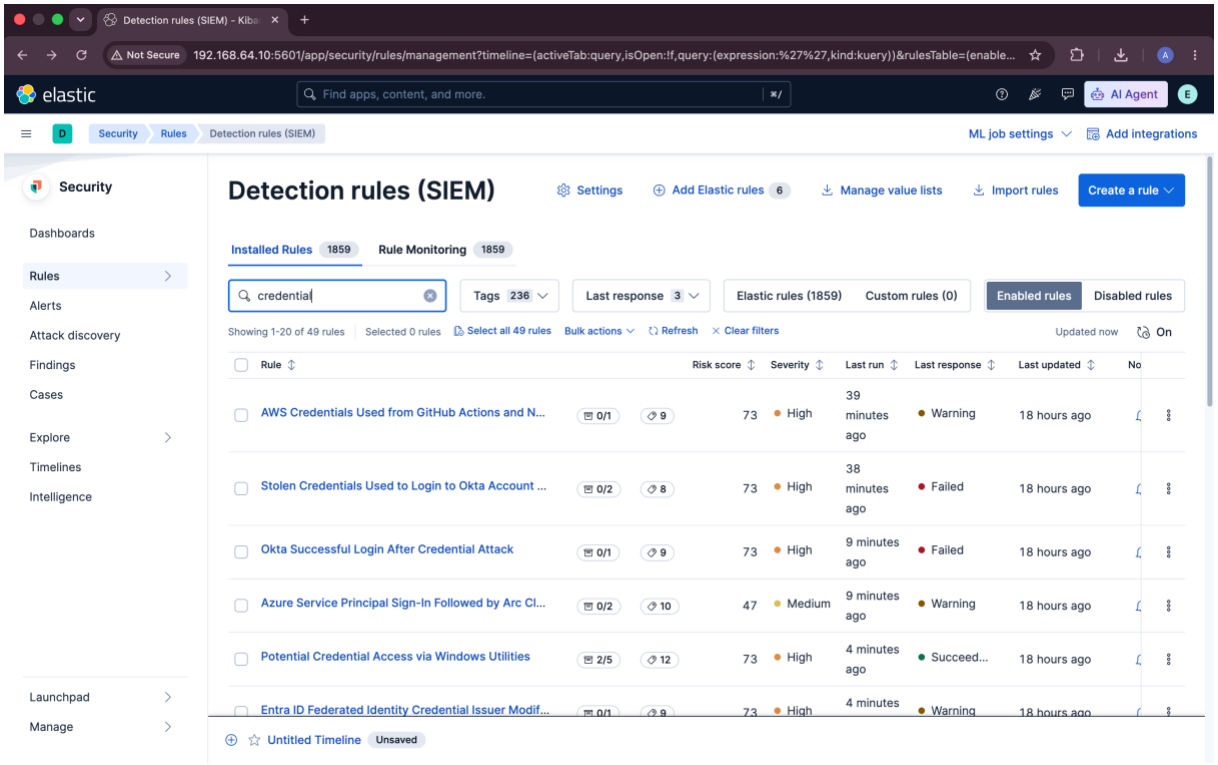


Figure 6. Credential-access and token-related rules enabled.

4.6 Registry rule coverage

Registry monitoring was necessary because registry changes are commonly used for persistence and configuration manipulation.

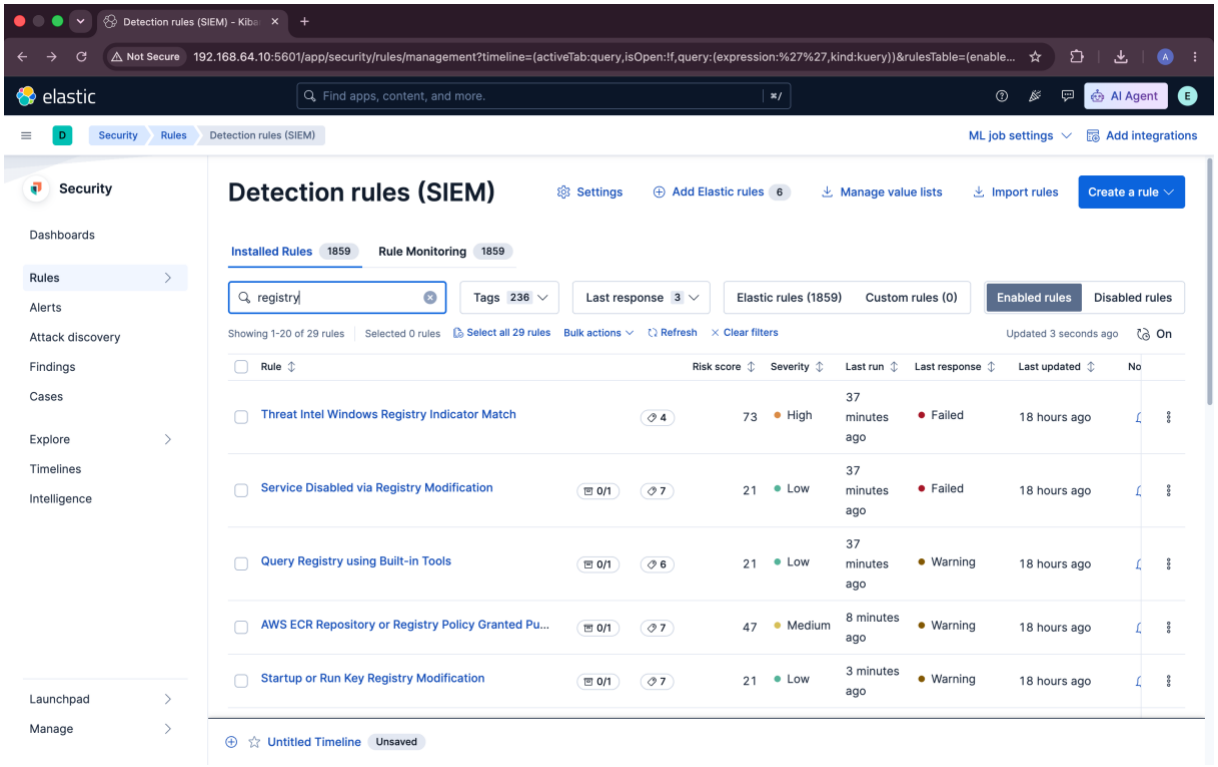


Figure 7. Registry-related detection rules enabled.

4.7 Process rule coverage

Process rules gave the lab visibility into process creation, suspicious parent-child relationships and execution patterns.

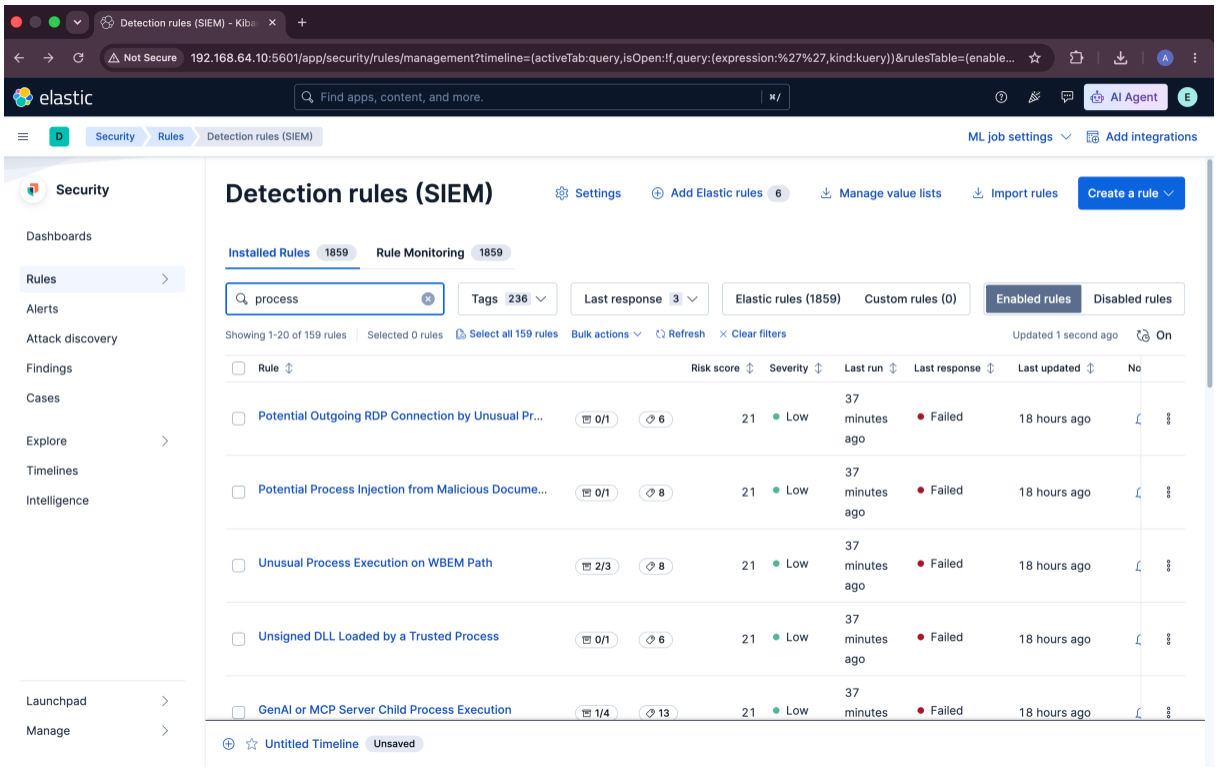


Figure 8. Process execution and injection-related rules enabled.

5. Endpoint onboarding and telemetry

The Windows endpoint was enrolled through Elastic Fleet. Fleet provided central visibility of agent health, policy assignment and check-in status. The Windows integration and Sysmon telemetry supplied the event detail needed for useful investigations.

5.1 Fleet health

A healthy Fleet view confirmed that the management server and endpoint agent were connected and reporting. This was the essential proof that the data collection path was available.

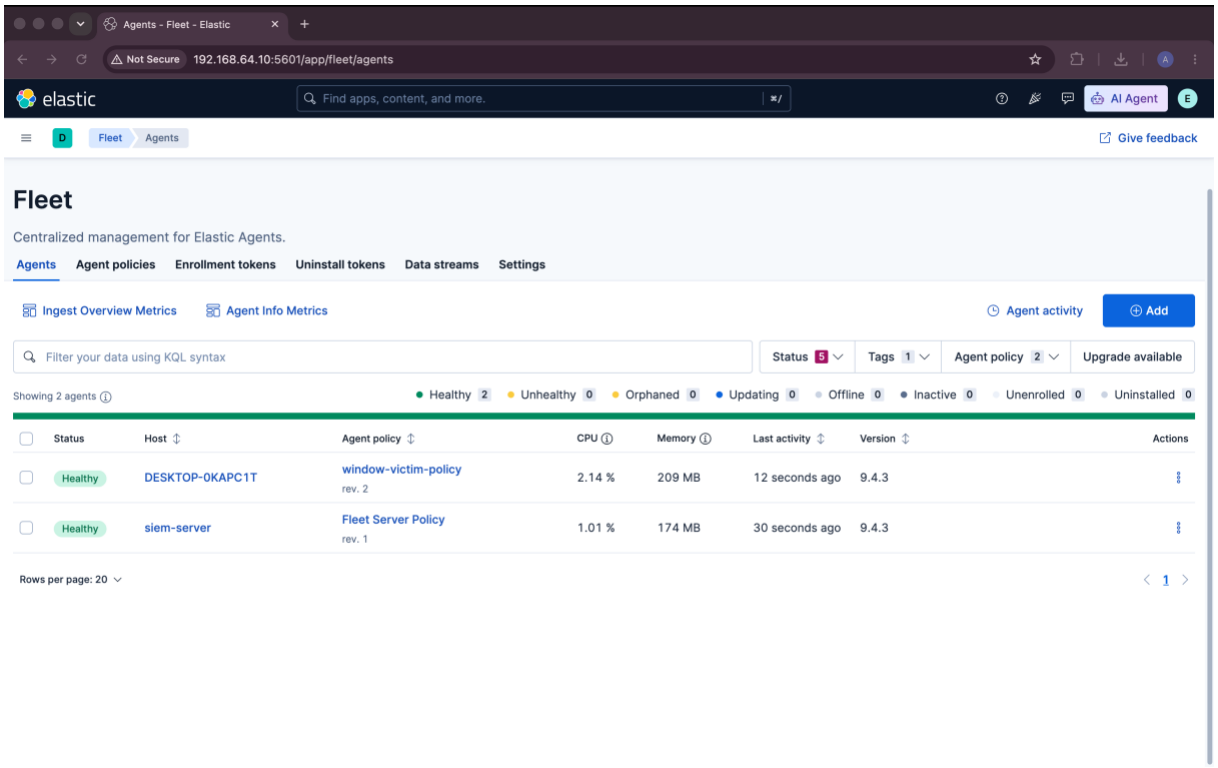


Figure 9. Fleet showing the Windows endpoint and Fleet Server in a healthy state.

5.2 Windows agent verification

The endpoint-side verification confirmed that the agent was installed, connected and capable of forwarding telemetry.

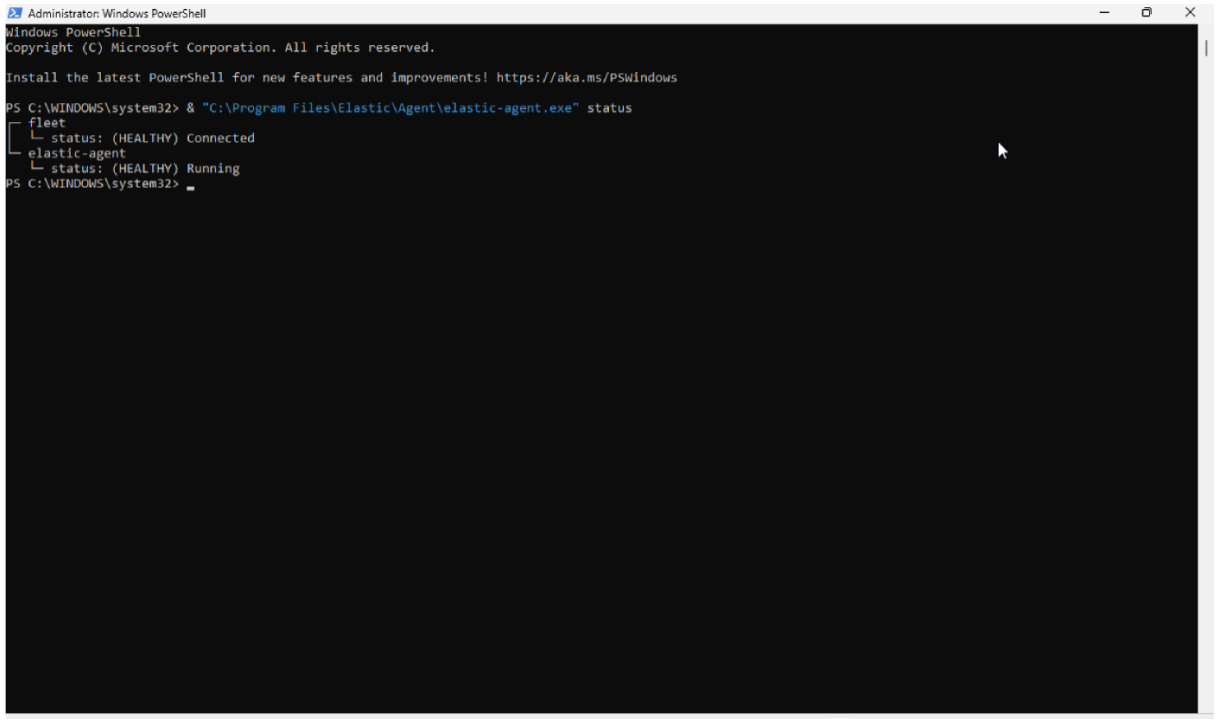


Figure 10. Windows endpoint confirming Elastic Agent activity.

5.3 Authentication events

Windows event IDs 4624 and 4625 were used to validate successful and failed authentication logging. These events are fundamental for account misuse and brute-force investigations.

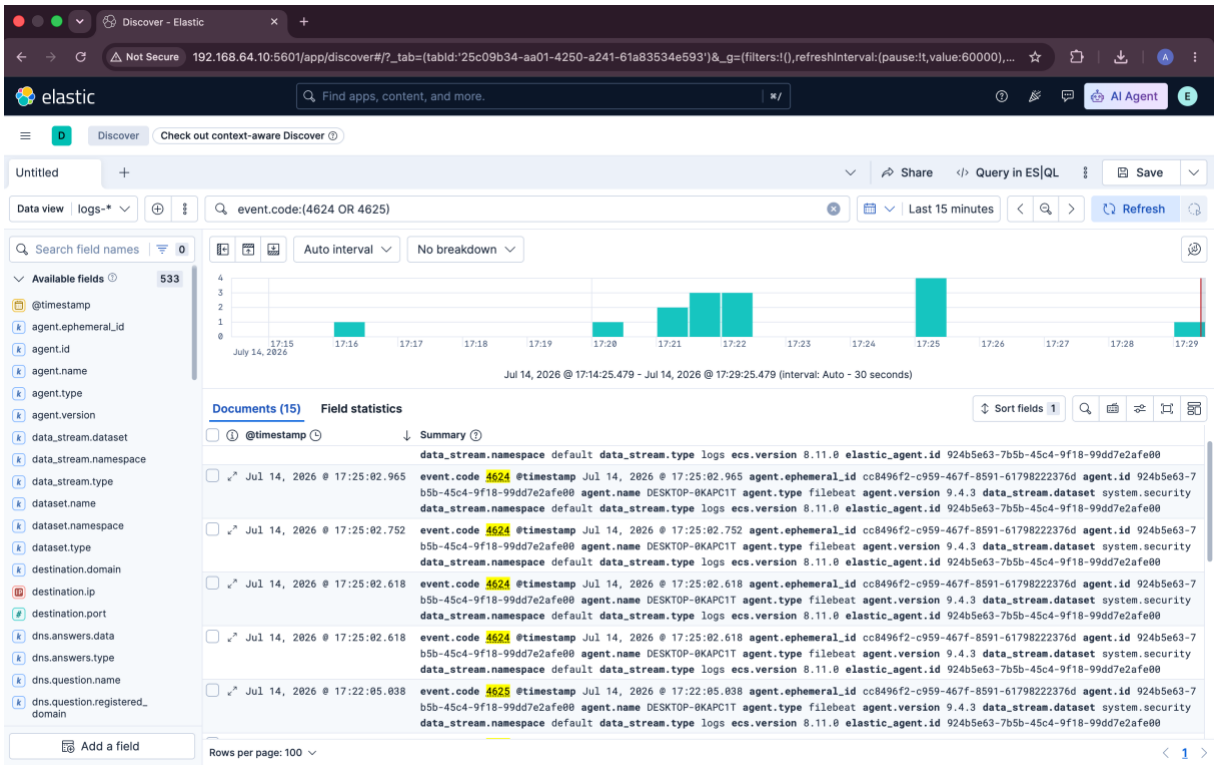


Figure 11. Successful and failed Windows authentication events visible in Elastic.

5.4 Failed authentication test

Controlled failed logons were generated to prove that the endpoint, agent, SIEM and detection workflow could carry authentication failures through the pipeline.

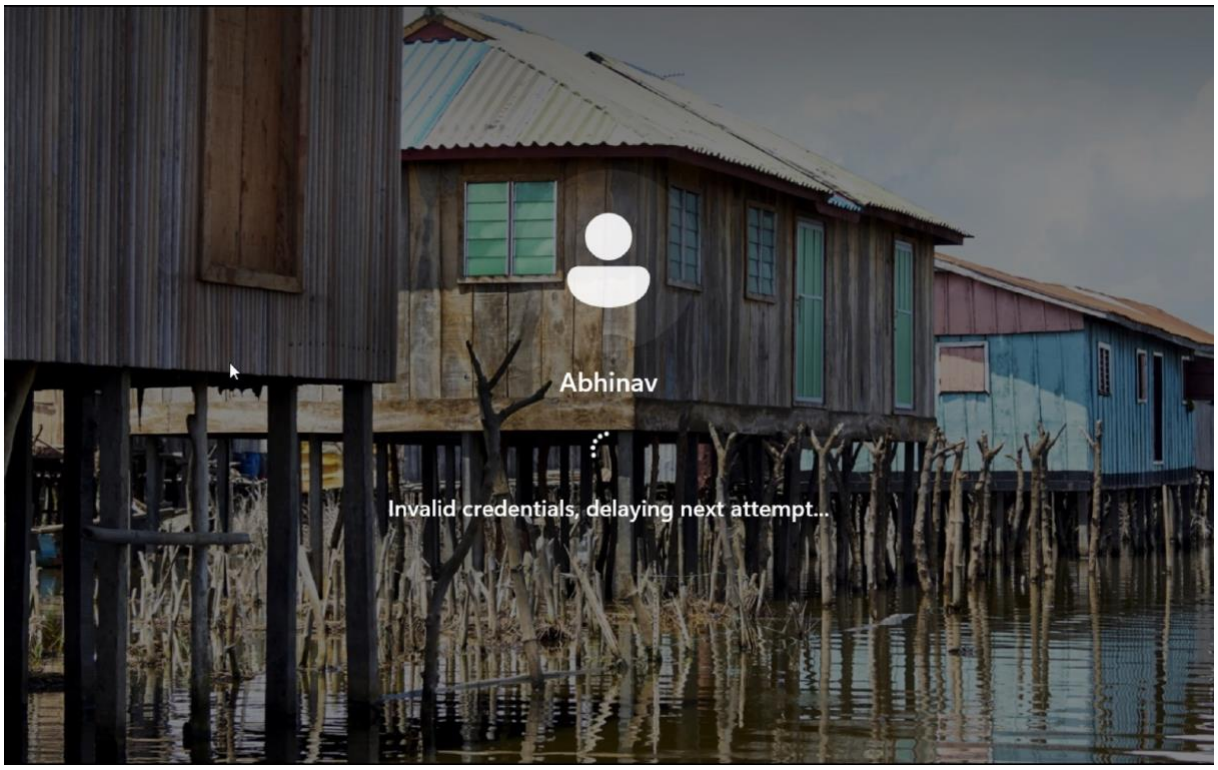


Figure 12. Controlled failed authentication activity generated on the Windows endpoint.

5.5 PowerShell telemetry, first view

PowerShell telemetry was inspected directly to confirm that command activity was being captured before relying on detection rules.

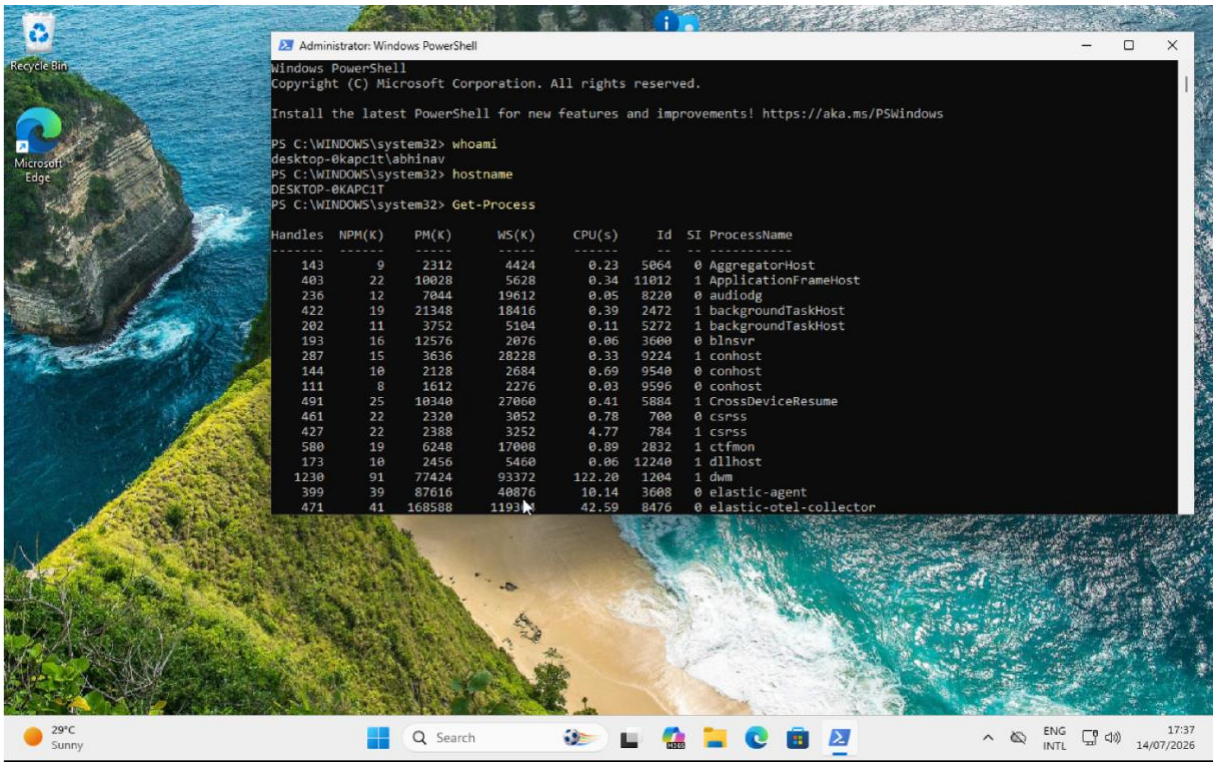


Figure 13. PowerShell telemetry collected from the endpoint.

5.6 PowerShell telemetry, second view

The second view provided additional confirmation that the event fields required for later correlation and evidence presentation were present.

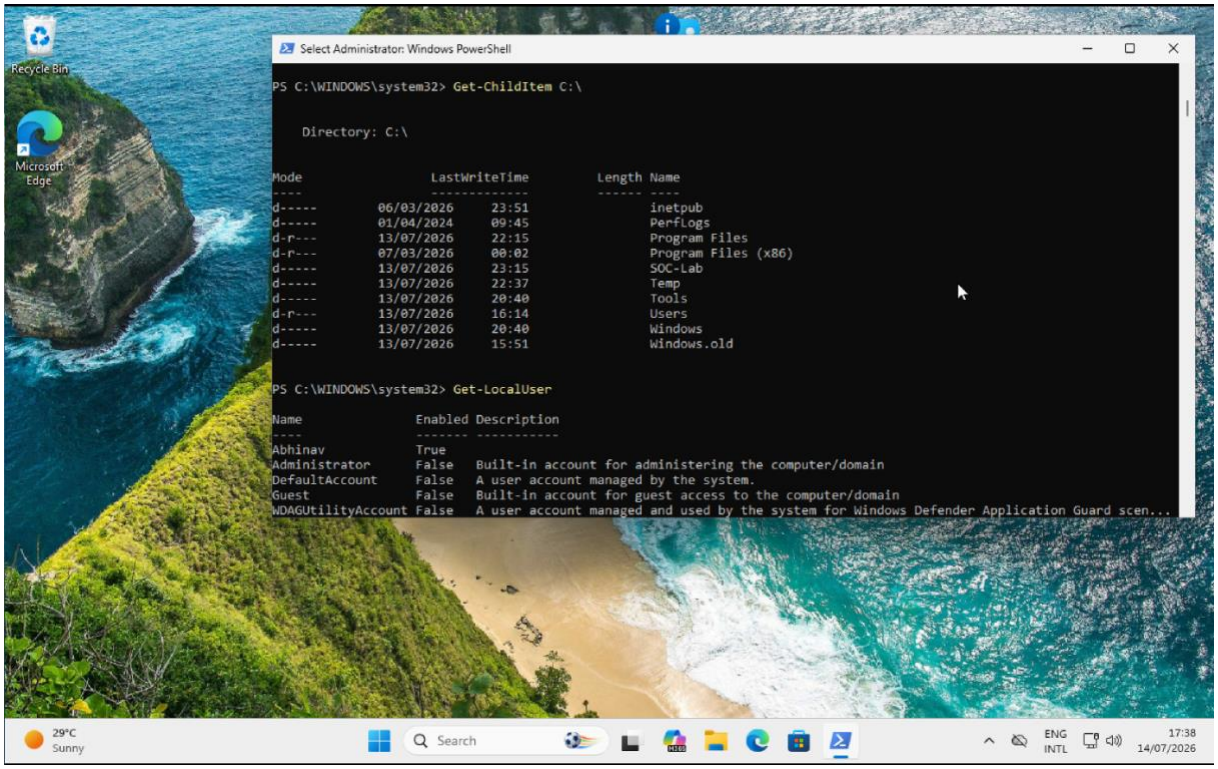


Figure 14. Additional PowerShell event detail available for investigation.

5.7 Process event telemetry

Process creation telemetry links a command to its executable, user, host and parent process. These relationships are important for building an attack chain.

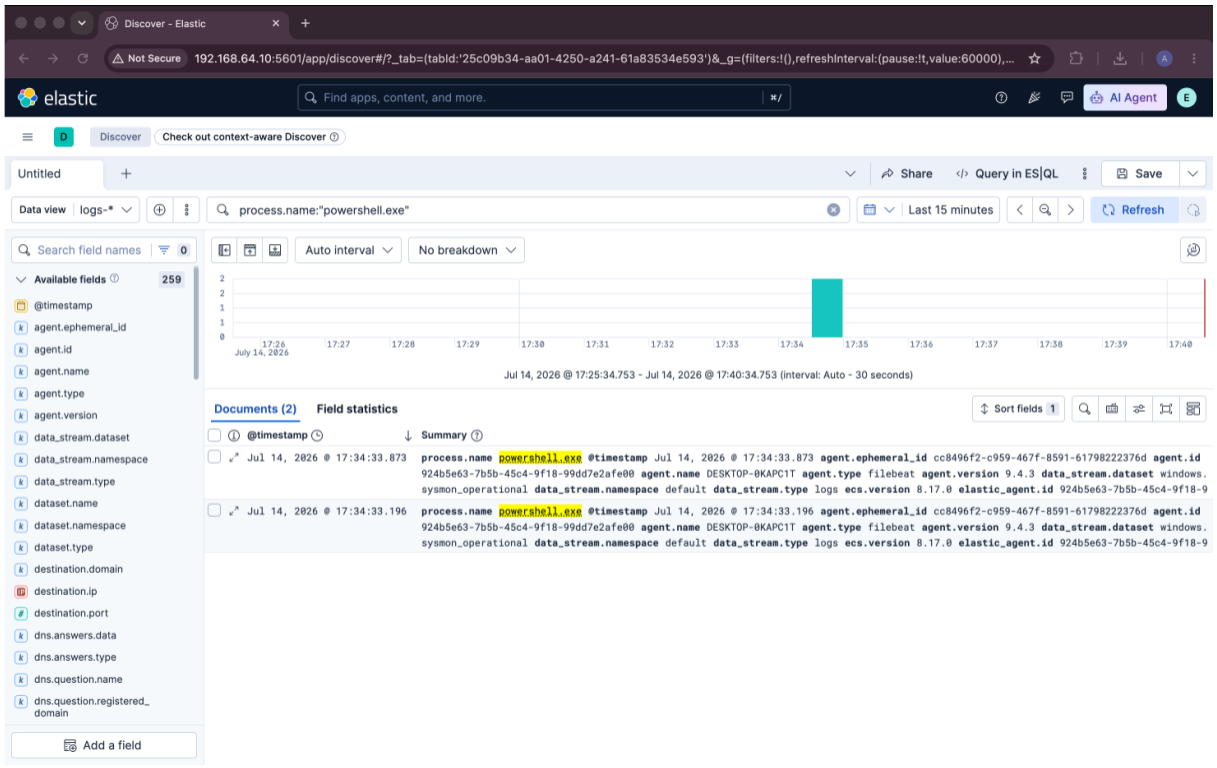


Figure 15. Process execution events associated with PowerShell activity.

6. Event generation and validation in Discover

After the endpoint and agent were confirmed healthy, controlled activity was generated across several Windows behaviours. Elastic Discover was used to verify the underlying events before examining alerts. This order matters: a detection cannot be trusted if the raw telemetry that should support it is missing.

6.1 PowerShell event search

The Discover query confirmed the presence and timing of PowerShell events generated during the test.

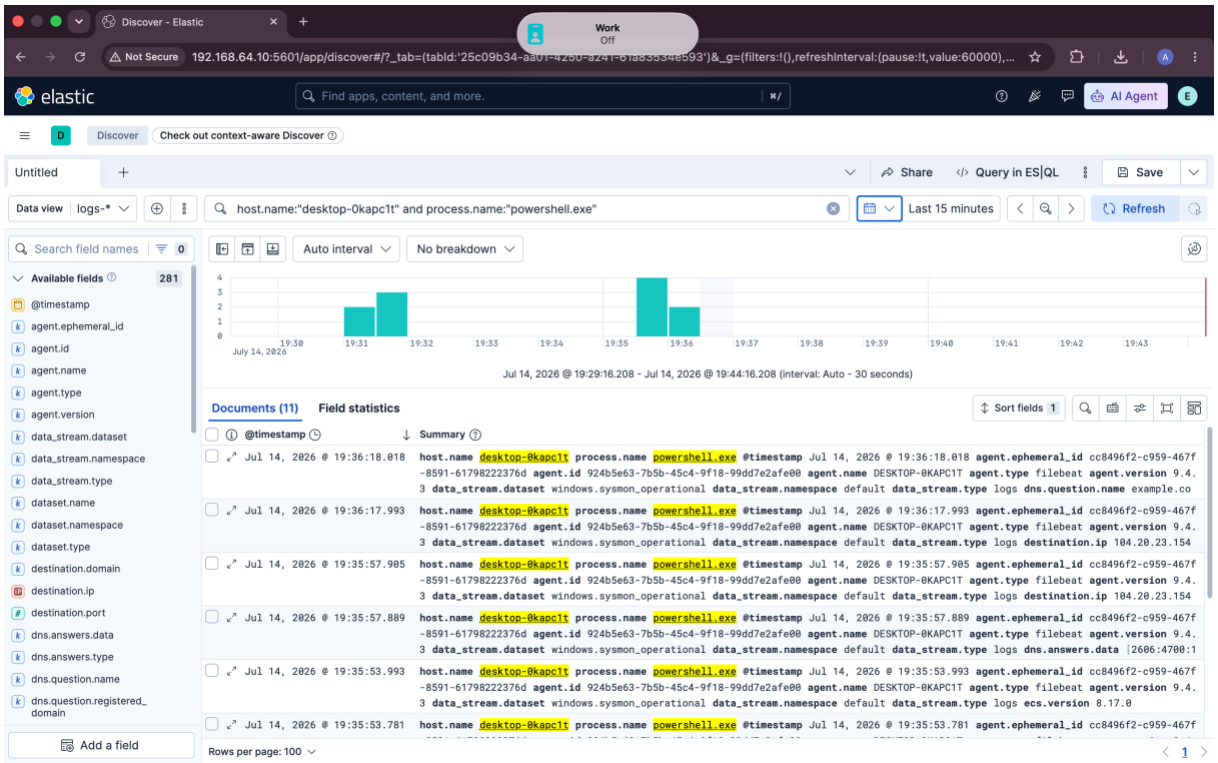


Figure 16. PowerShell activity located in Elastic Discover.

6.2 Curl event search

Curl activity was generated to create an observable file-transfer or network-access pattern.

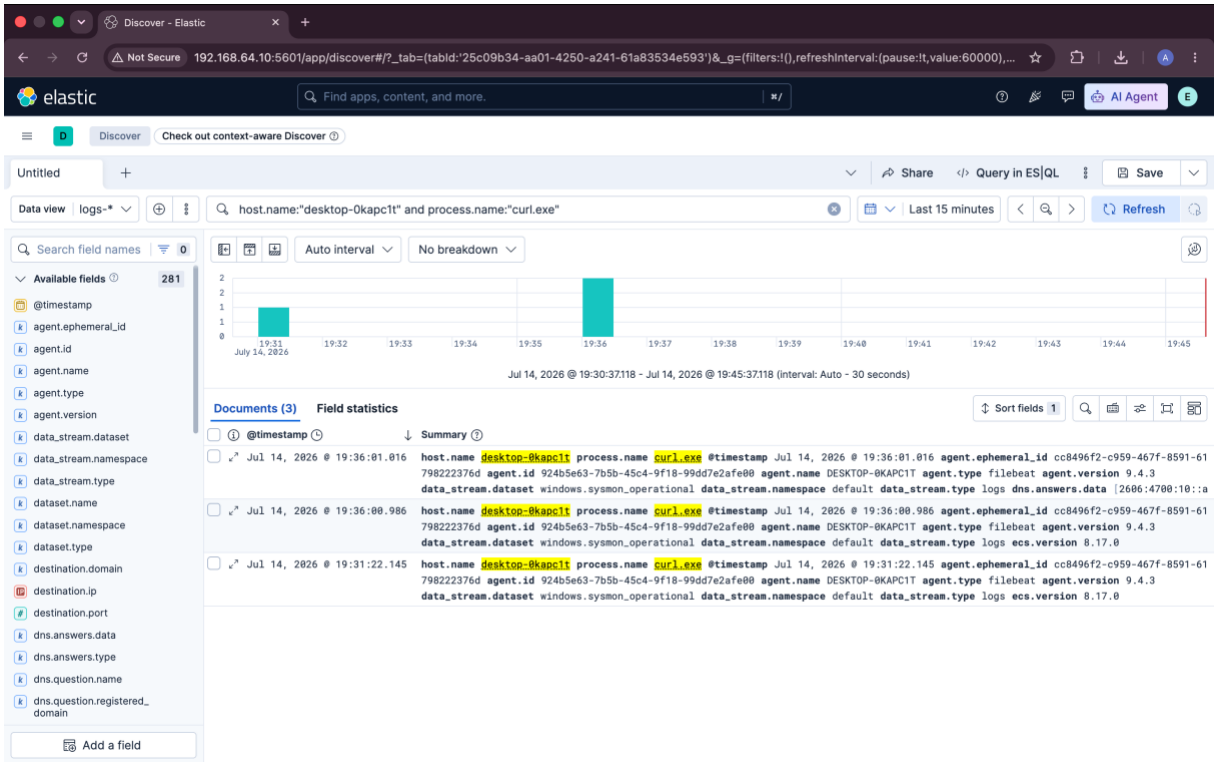


Figure 17. Curl-related process and network activity in Discover.

6.3 Certutil event search

Certutil is a signed Windows utility that can be used legitimately or abused for transfer and decoding. Its activity provided another useful test case.

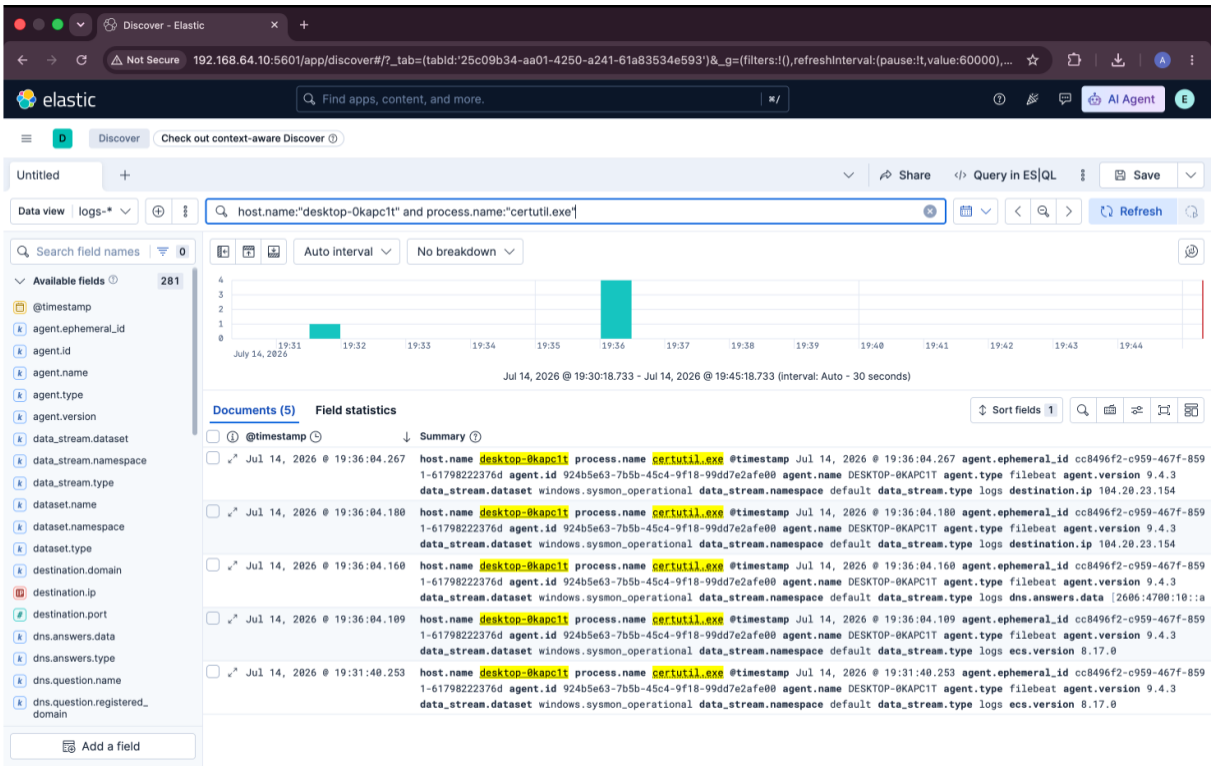


Figure 18. Certutil activity visible in the collected telemetry.

6.4 DNS event search

DNS data was checked because network investigations depend on linking endpoint processes to queried domains.

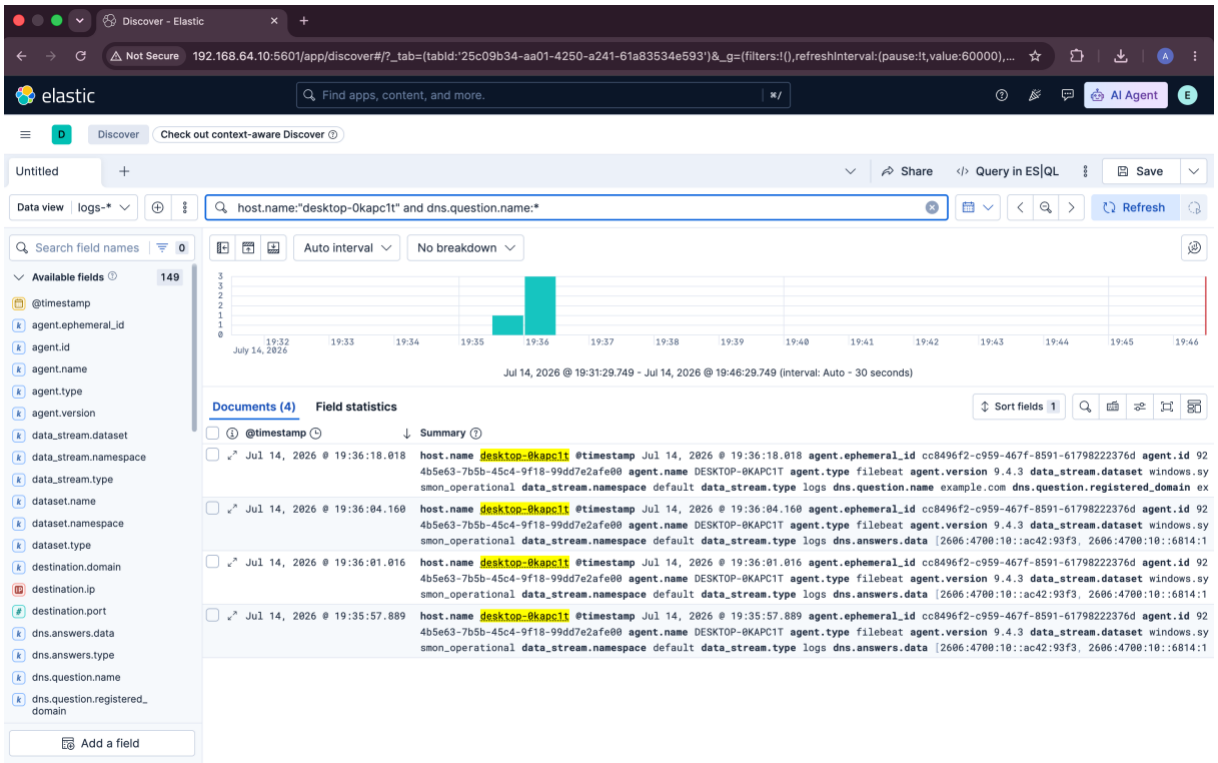


Figure 19. DNS resolution activity visible in Discover.

6.5 Scheduled task event search

Scheduled tasks are a common persistence mechanism. Creating and removing a controlled task confirmed that this behaviour was visible.

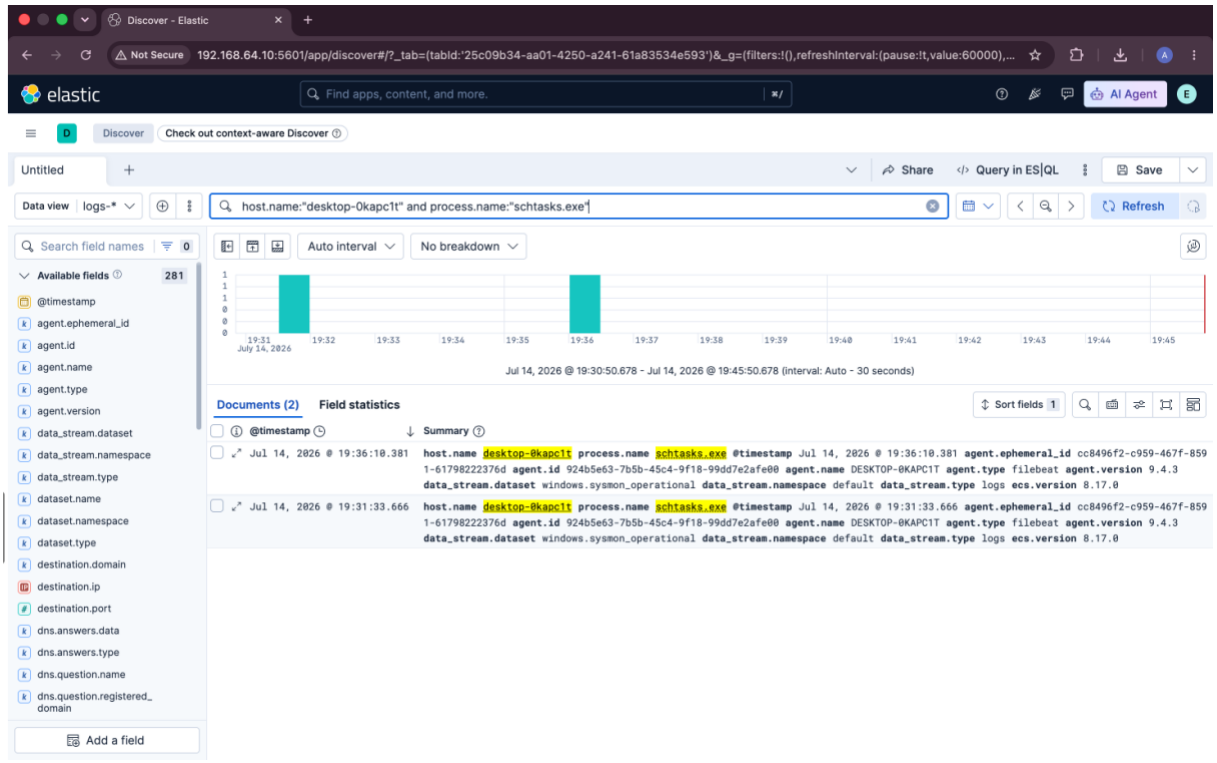


Figure 20. Scheduled-task activity recorded by the endpoint.

6.6 Registry event search

The registry test confirmed that configuration and persistence-related changes were collected with sufficient context.

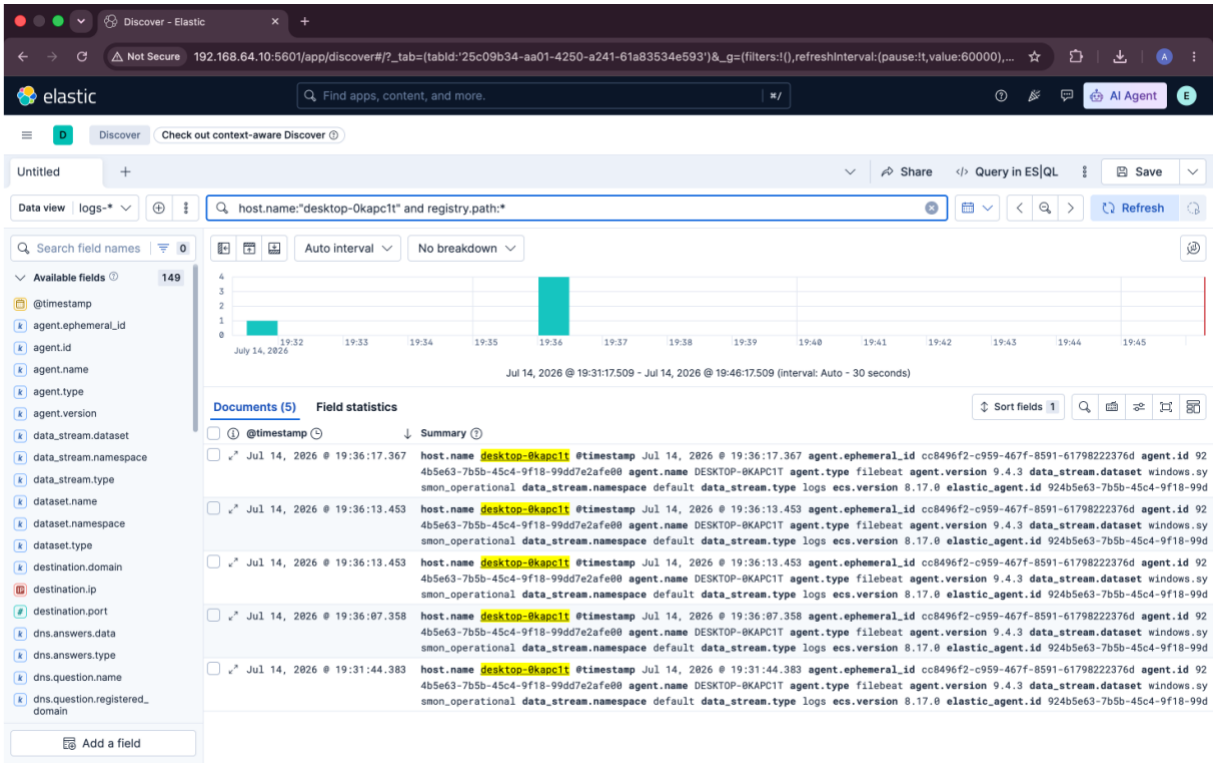


Figure 21. Registry modification activity in Discover.

6.7 Service creation event search

Service creation was included because services are frequently used for persistence and execution.

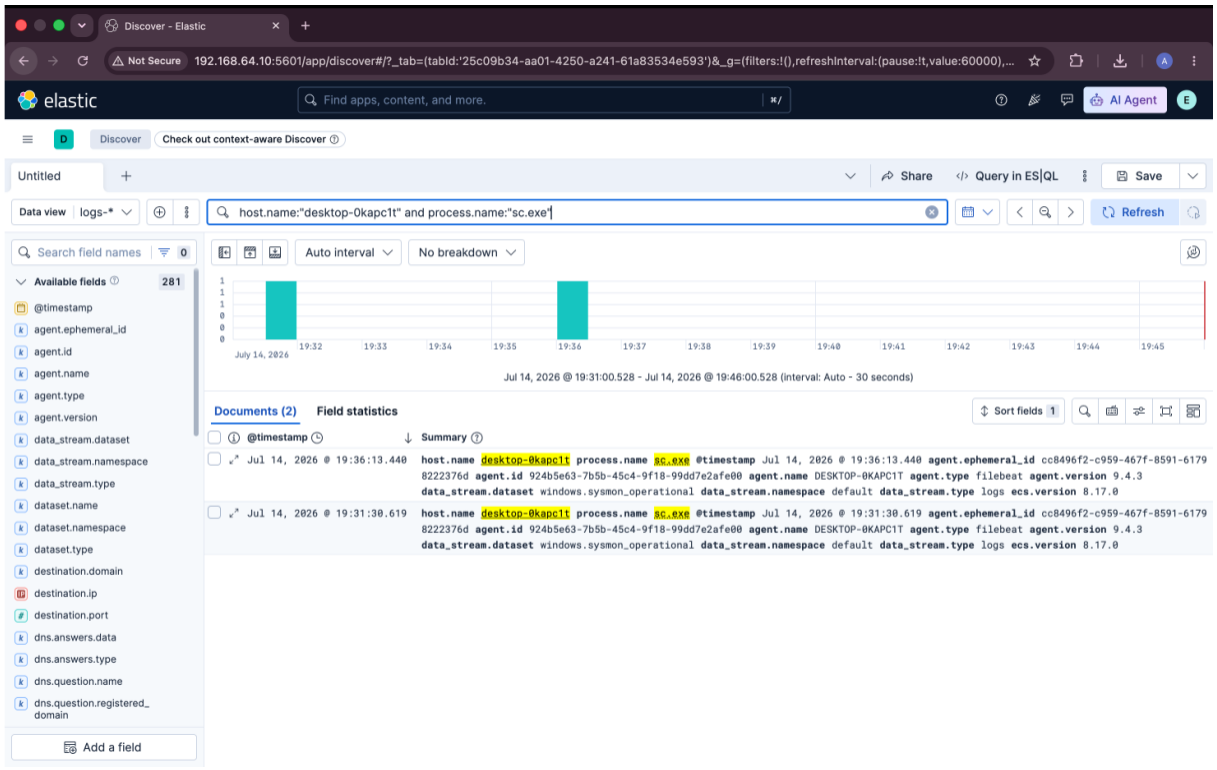


Figure 22. Windows service creation activity visible in Elastic.

7. Detection and manual SOC investigation

Once the raw events were verified, the Elastic Security Alerts interface was used to confirm that the enabled detection rules were producing analyst-facing alerts. The investigation then moved through the normal SOC workflow: alert review, detailed inspection, timeline creation and case recording.

7.1 PowerShell detection alert

This alert demonstrated that the rule engine successfully converted endpoint telemetry into a security detection.

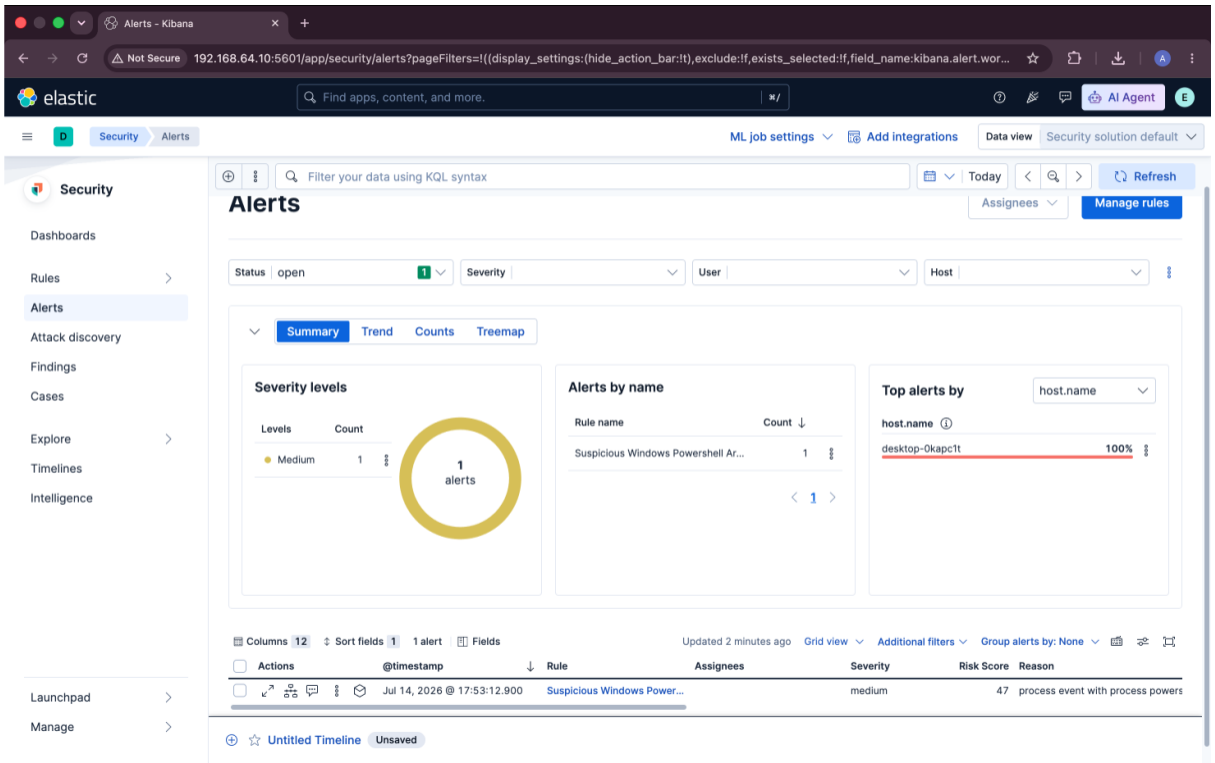


Figure 23. Elastic Security alert generated from suspicious PowerShell activity.

7.2 Combined PowerShell and curl detection

The combined view provided the first clear indication that multiple related activities could form a larger investigation rather than separate alerts.

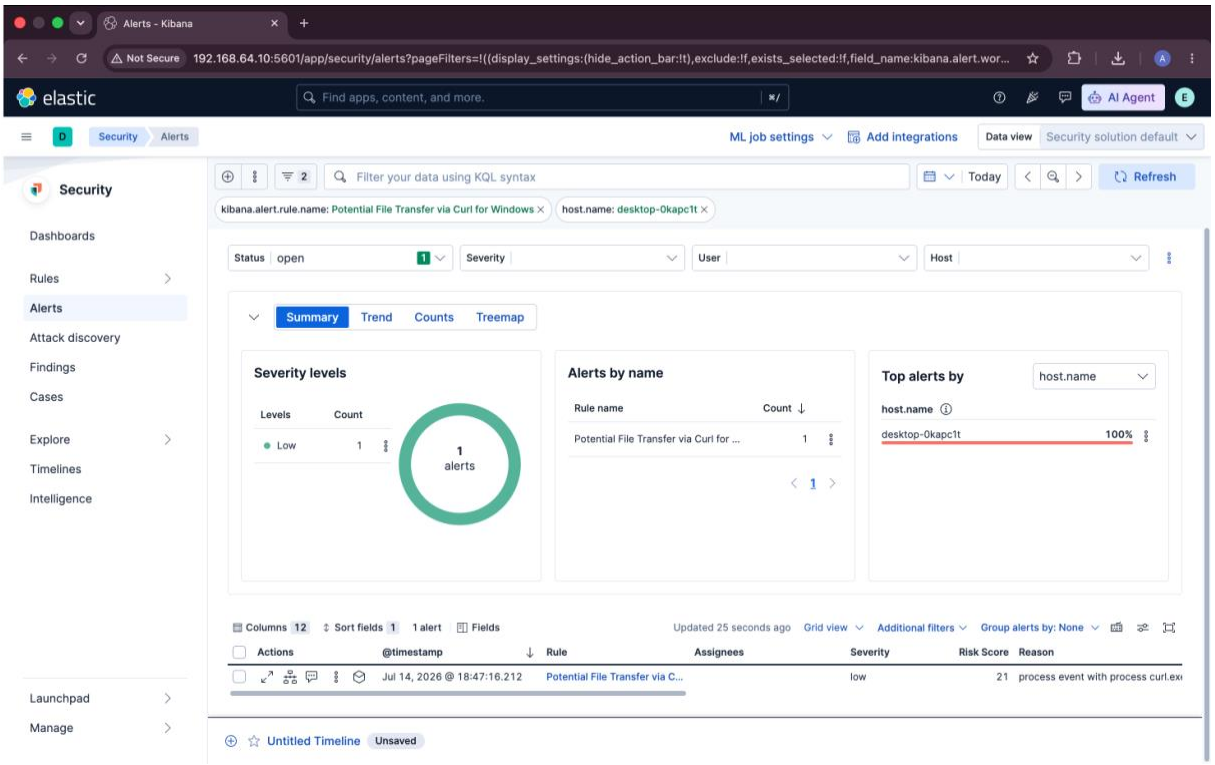


Figure 24. Related PowerShell and curl detections visible together.

7.3 Alert investigation panel

The alert detail panel was used to inspect the rule name, severity, timestamp, host, user and supporting event data.

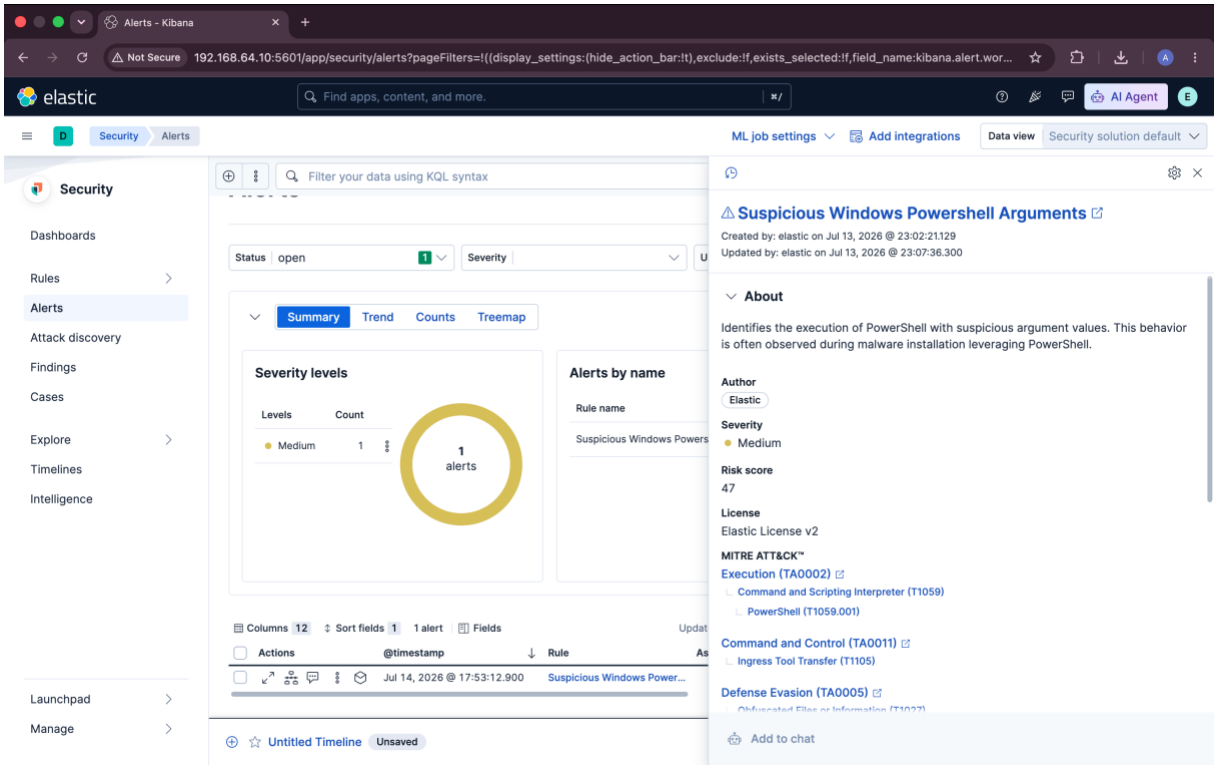


Figure 25. Detailed Elastic alert view with rule and event information.

7.4 Alert overview

The alert overview showed the accumulated results of the test sequence and the distribution of rule activity.

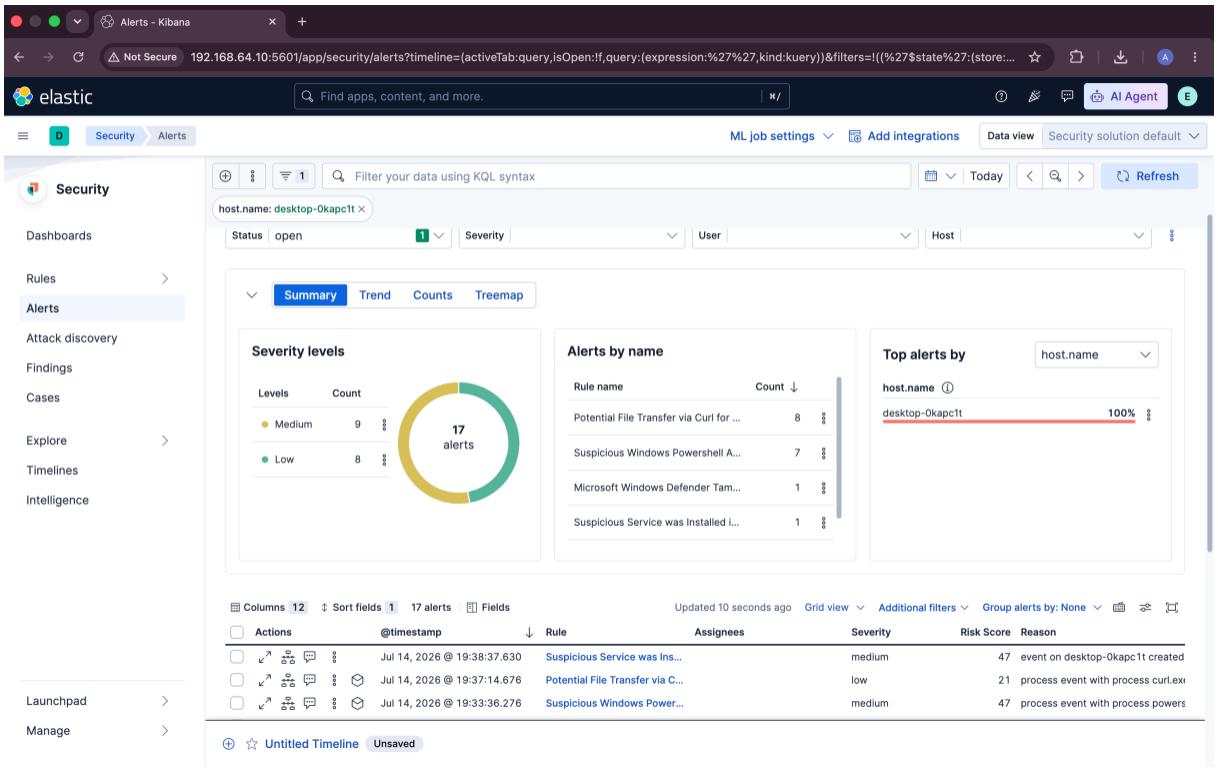


Figure 26. Elastic Security alert summary after controlled activity generation.

7.5 Security dashboard

The dashboard provided a higher-level summary of alert counts, severity and the most active endpoint.

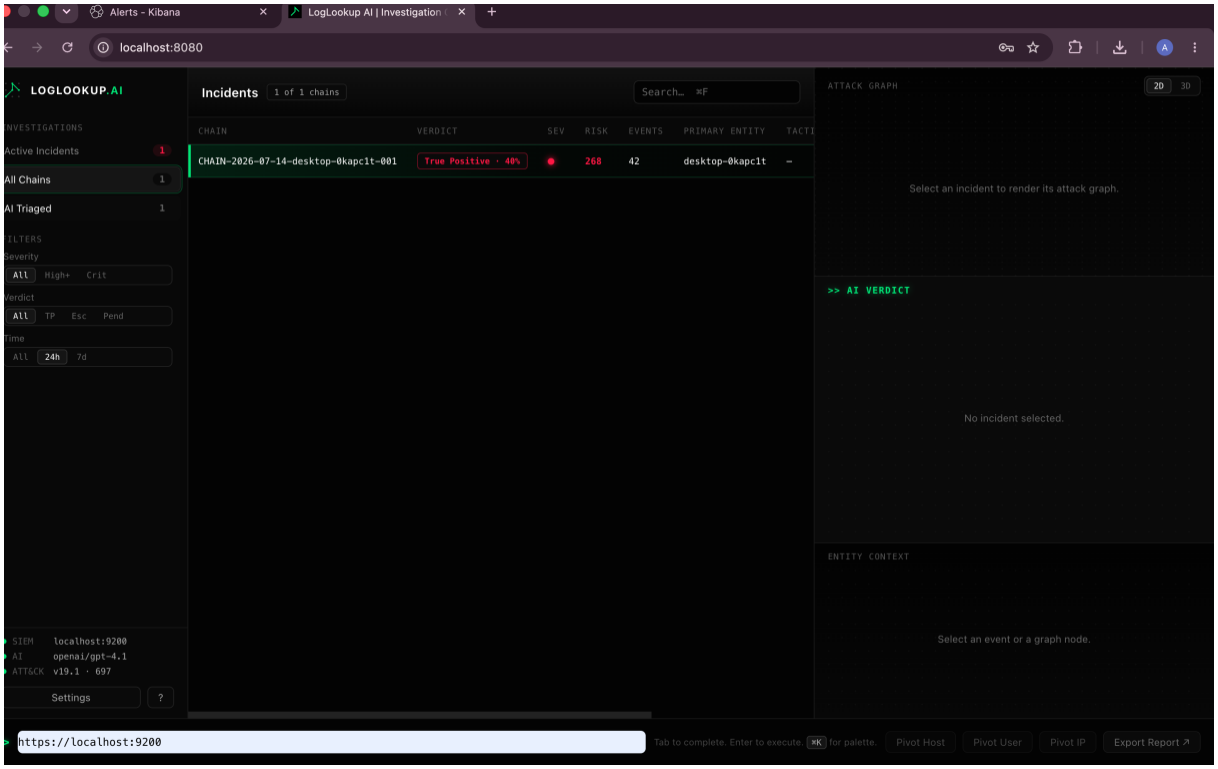


Figure 27. Elastic Security dashboard summarising the generated detections.

7.6 Timeline creation

Timeline was used to bring related events into one analyst workspace and review their sequence.

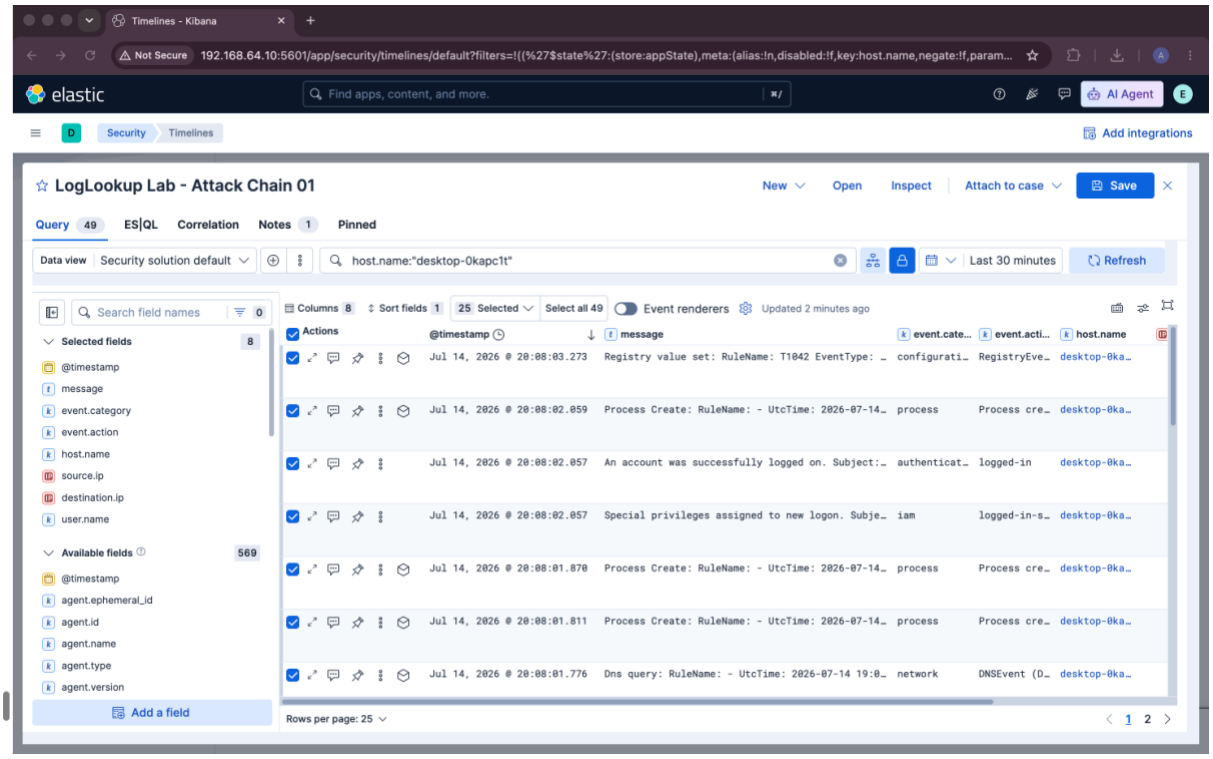


Figure 28. Elastic Timeline used to organise events for manual investigation.

7.7 Case record

A case was created to preserve the investigation summary, associated alert and analyst notes.

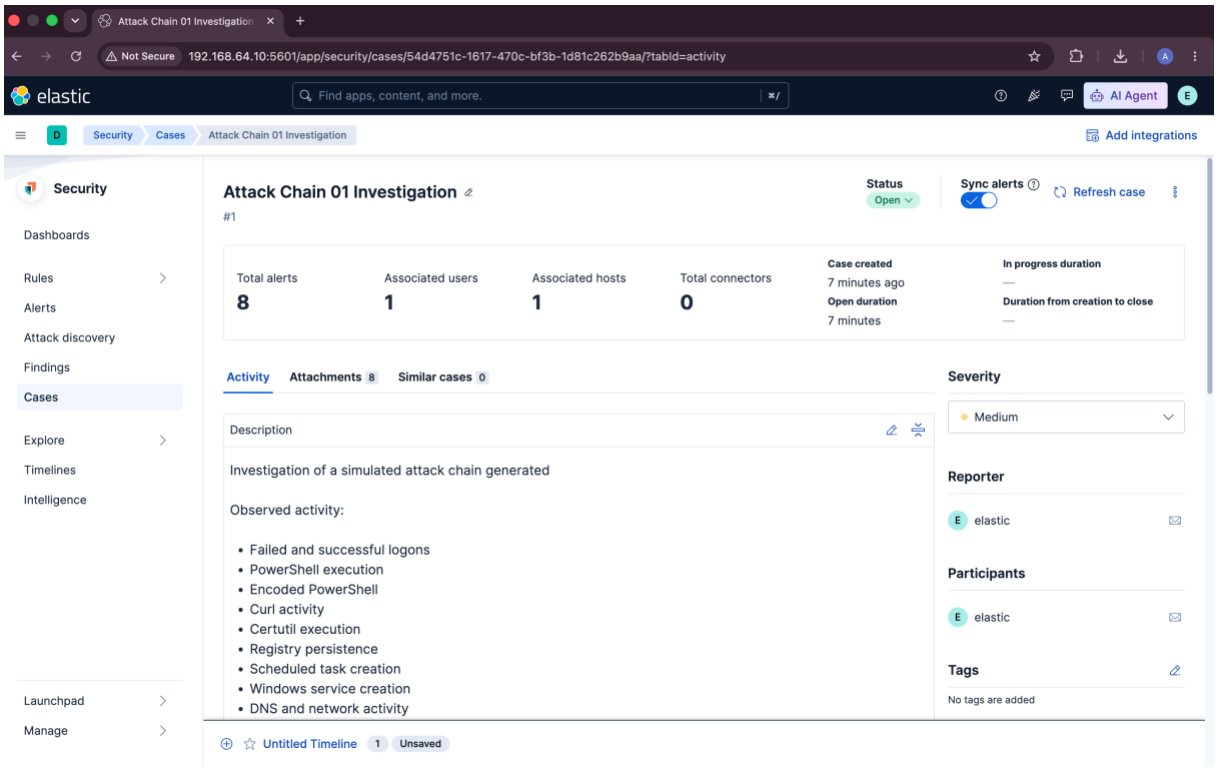


Figure 29. Elastic case created for the attack-chain investigation.

7.8 Case report

The case report captured the manual SOC interpretation of the generated activity and served as a comparison point for LogLookup AI.

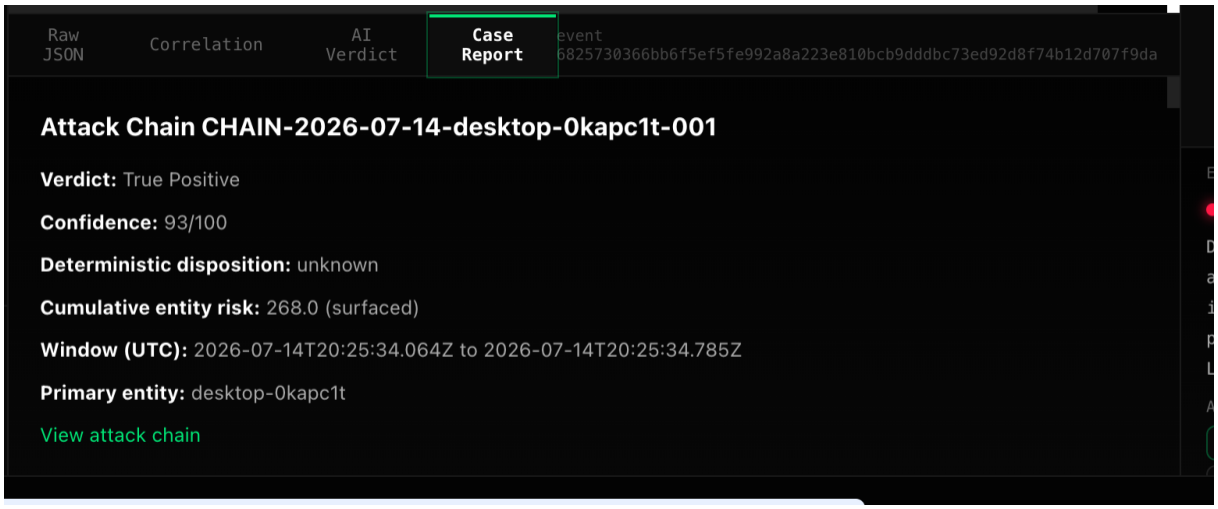


Figure 30. Case report documenting the manually investigated activity.

8. LogLookup AI installation and configuration

After the Elastic lab had been fully validated, LogLookup AI was installed on the Ubuntu SIEM server from the project repository. The application was installed into a dedicated Python environment, registered as a user service and opened through an SSH tunnel from macOS. The onboarding wizard was then used to connect the application to Elasticsearch and configure its analysis provider.

8.1 Welcome and first launch

The welcome screen introduced the configuration sequence and confirmed that the web application was reachable.

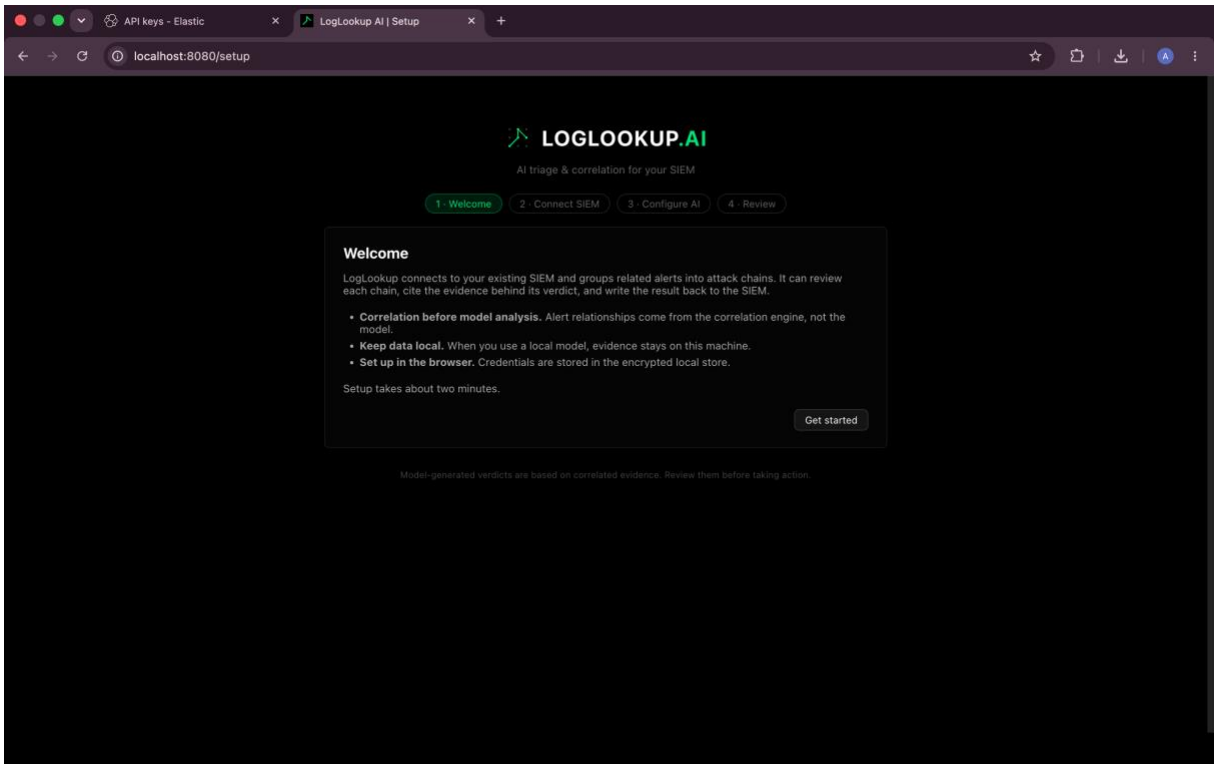


Figure 31. LogLookup AI welcome page displayed on first launch.

8.2 SIEM connection

The SIEM step stored the Elasticsearch URL, API key, alert index, polling interval and minimum severity.

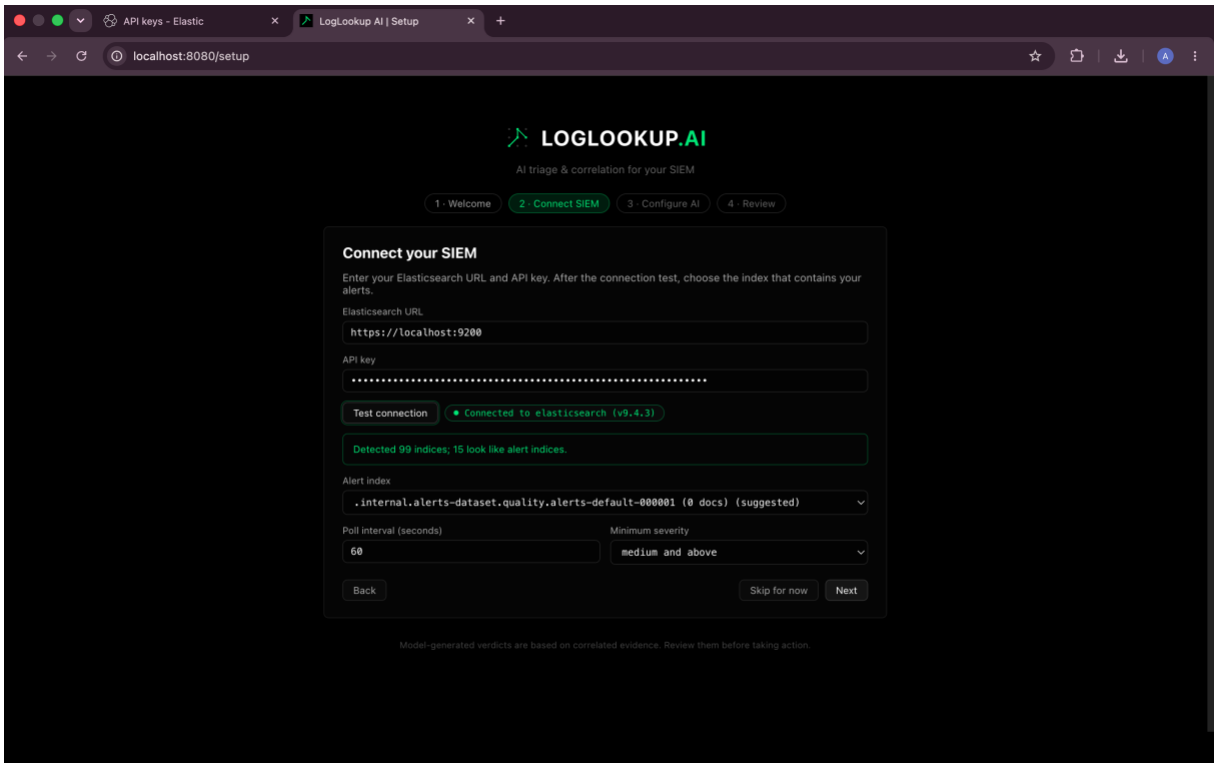


Figure 32. Elasticsearch connection tested and alert index selected.

8.3 AI provider configuration

The AI configuration step selected the provider and model and enabled redaction before evidence was sent.

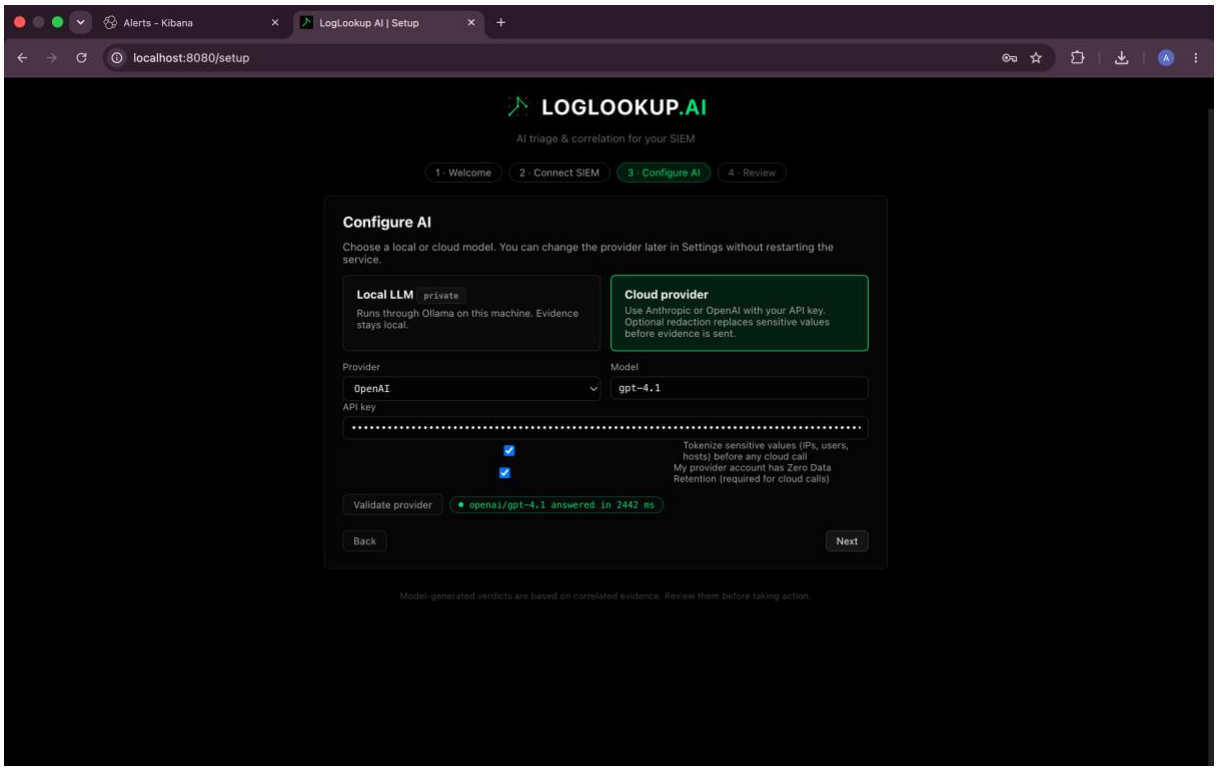


Figure 33. Cloud AI provider configured and validated.

8.4 Review and launch

The review page confirmed that all required components were available before the service began normal polling.

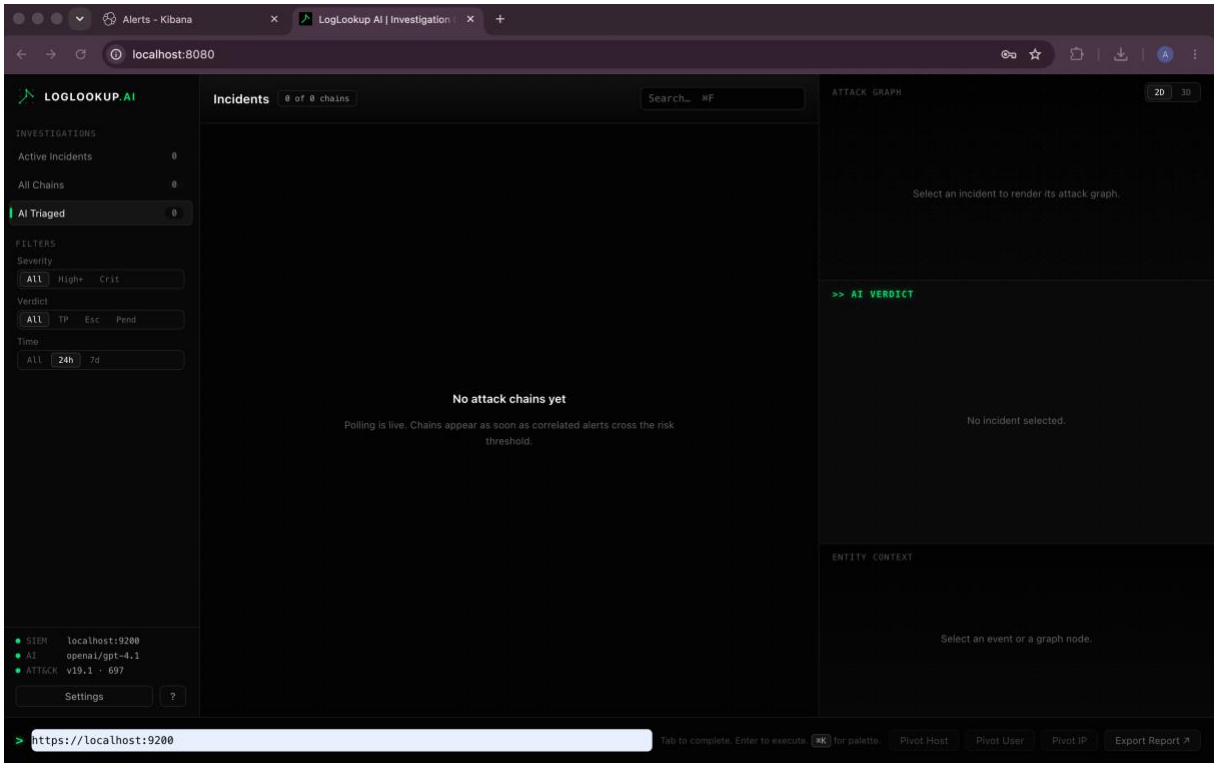


Figure 34. Final review page confirming SIEM, AI provider and ATT&CK knowledge-base status.

8.5 Elastic API key

A dedicated API key separated the application credentials from personal accounts and allowed controlled access to alerts and write-back functions.

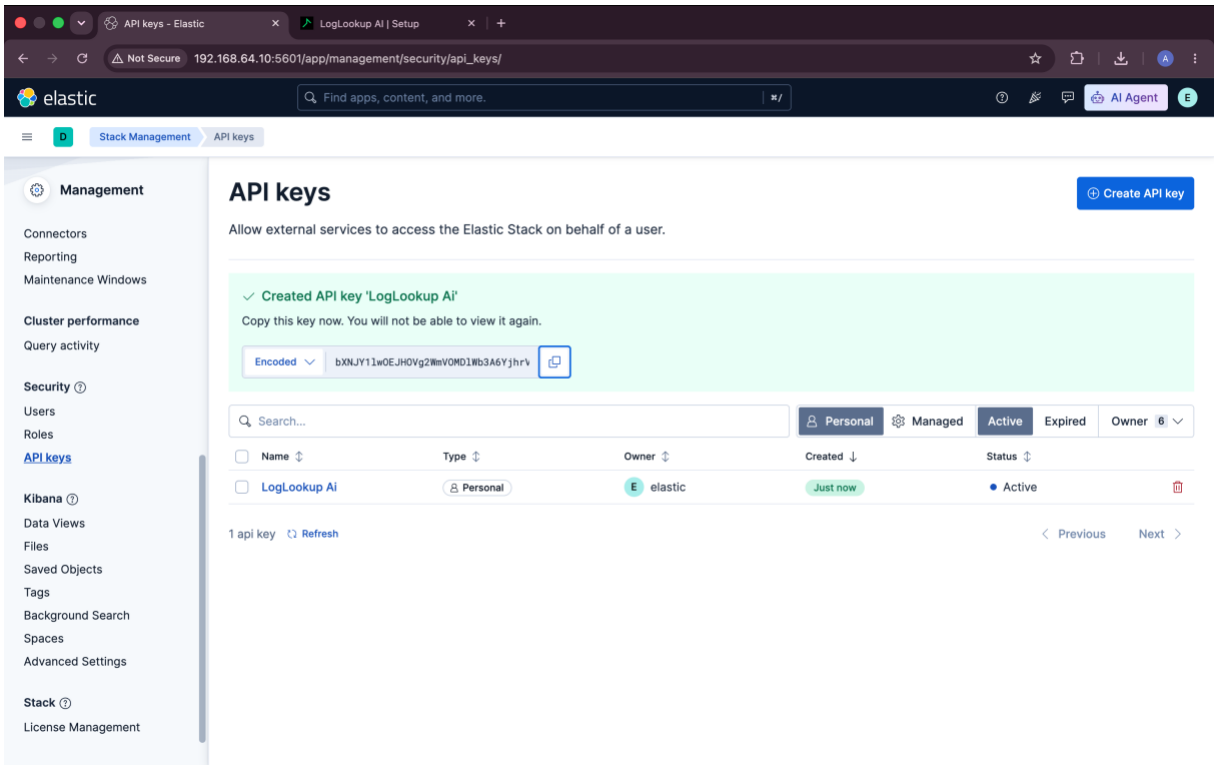


Figure 35. Dedicated Elastic API key created for LogLookup AI.

9. End-to-end LogLookup AI investigation

With configuration complete, LogLookup AI began polling the Elastic Security alert index. The application normalised the incoming alerts, linked shared entities, created a deterministic attack chain, calculated a cumulative risk score, submitted the chain for AI-assisted reasoning, validated the evidence references and wrote the completed result back to Elasticsearch.

9.1 Investigation dashboard

The dashboard confirmed that the application had created a single incident from the related alert set.

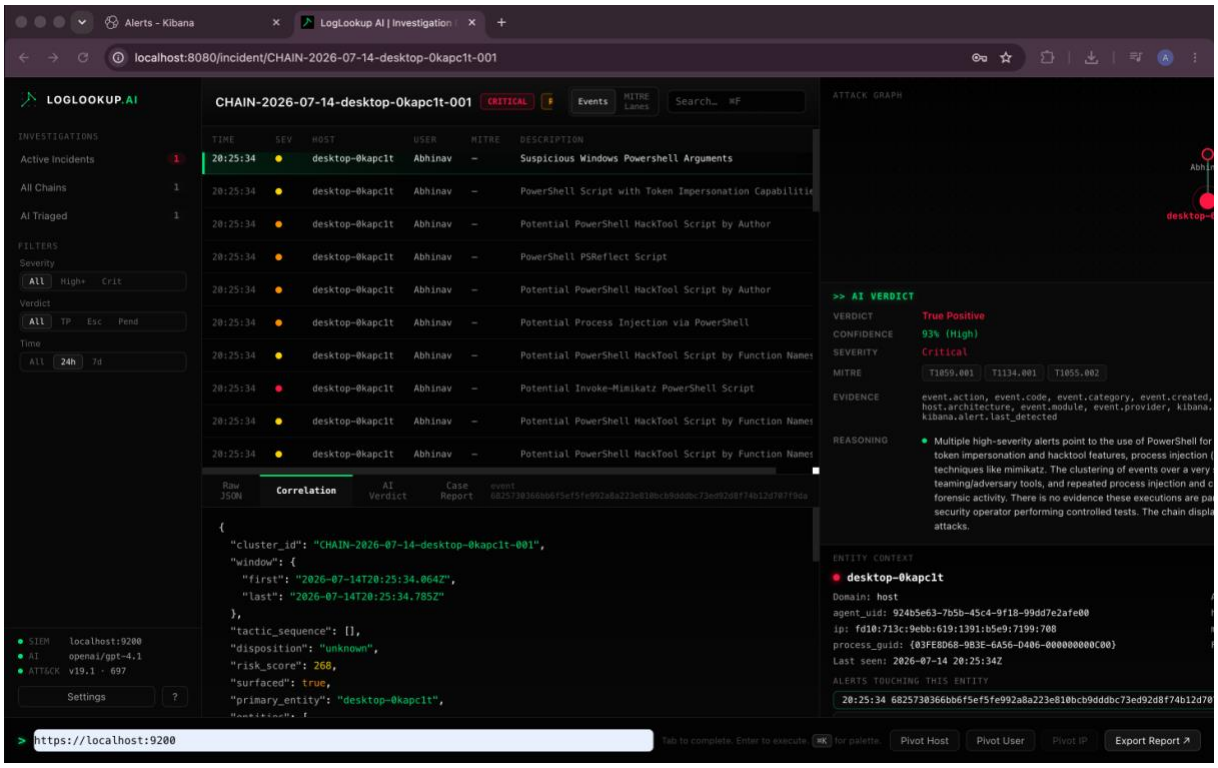


Figure 36. LogLookup AI dashboard showing the active correlated incident.

9.2 Final investigation workspace

The detailed workspace brought together event chronology, severity, risk, entity relationships and the AI-assisted conclusion.

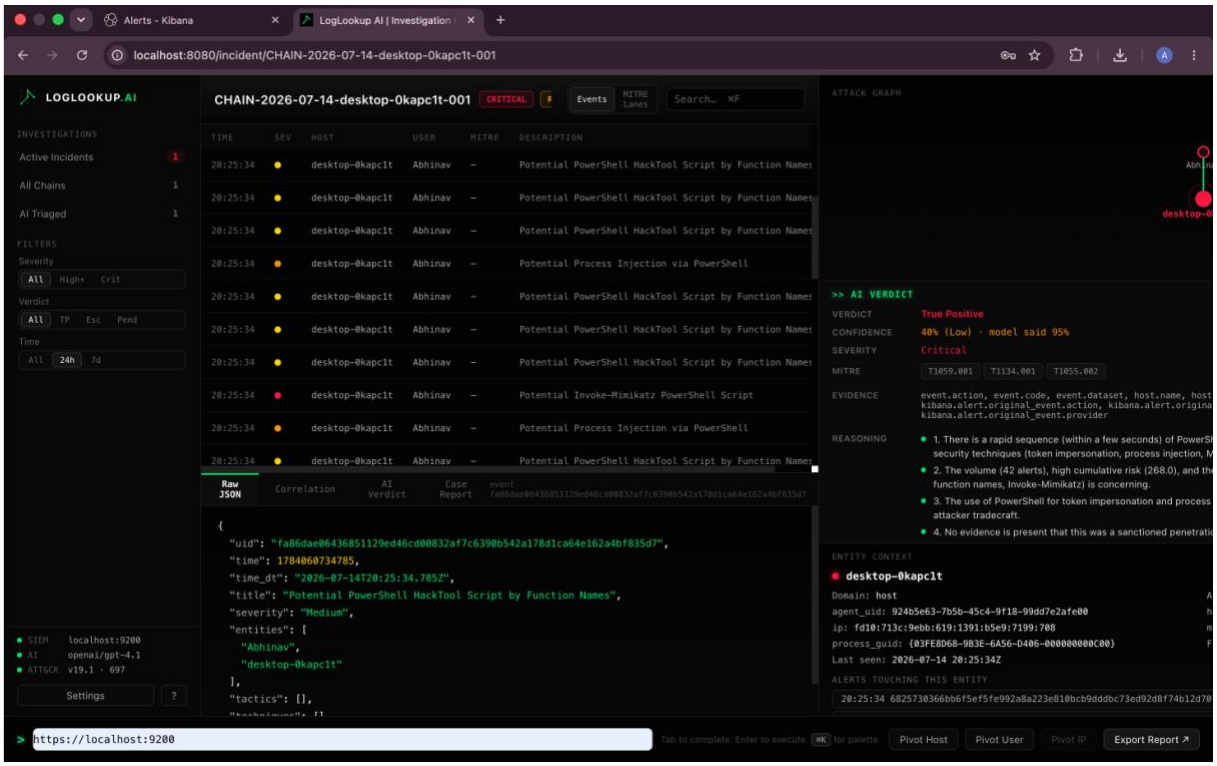


Figure 37. Completed investigation workspace with timeline, attack graph, verdict and entity context.

9.3 Raw JSON evidence

Raw JSON remained visible so the analyst could verify the exact fields and values behind the investigation.

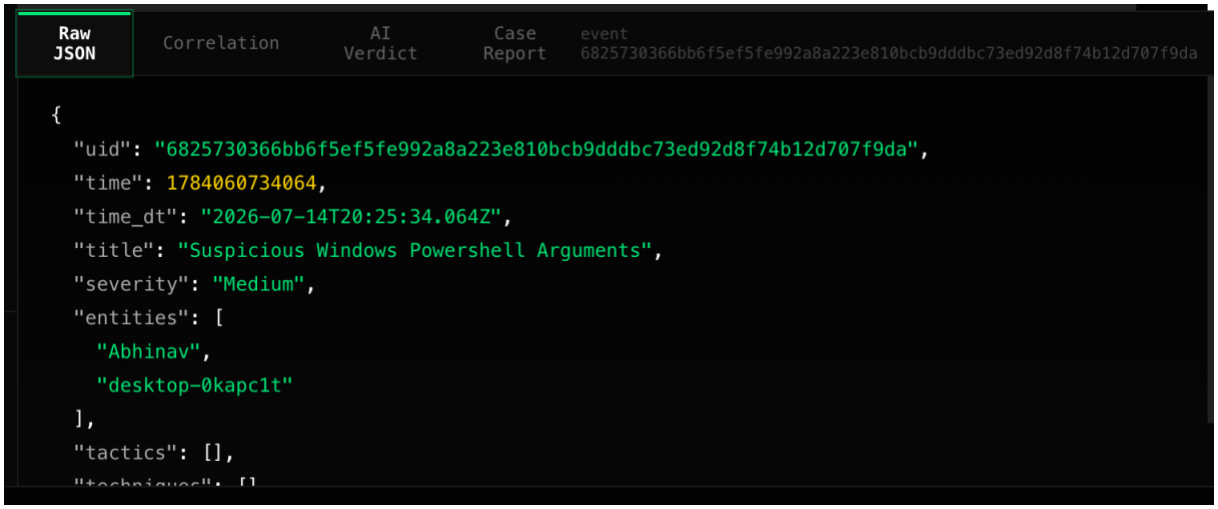


Figure 38. Raw normalised evidence available to the analyst.

9.4 Correlation output

The correlation panel exposed the cluster identifier, time window, primary entity, risk score and chain metadata.



Figure 39. Correlation metadata for the generated attack chain.

9.5 AI verdict

The AI verdict summarised the activity, assigned a true-positive assessment and presented the reasoning used to support the conclusion.



Figure 40. AI verdict and investigation reasoning generated from the correlated evidence.

10. Validation outcome

The completed lab demonstrated the full operational path from Windows activity to analyst investigation. The endpoint generated authentication, PowerShell, process, network, DNS, registry, scheduled-task and service telemetry. Elastic Agent and Sysmon carried that data to Elasticsearch. Detection rules created security alerts. The alerts were manually reviewed through Discover, Alerts, Timeline and Cases. LogLookup AI then consumed the same alert stream and produced a correlated incident with evidence, risk, entity context, attack-chain metadata and AI-assisted analysis.

The result is a practical portfolio demonstration of SIEM deployment, endpoint telemetry engineering, detection validation, SOC investigation and application integration. The screenshots in this report document each stage rather than only showing the final interface.

11. Conclusion

The project moved from an empty virtual laboratory to a working Elastic Security environment and then to a complete LogLookup AI investigation workflow. The work included infrastructure preparation, service installation, Fleet enrolment, Windows telemetry, Sysmon integration, detection-rule configuration, controlled activity generation, manual alert investigation, case creation, API-key configuration, application installation and final end-to-end validation.

The final investigation demonstrated that LogLookup AI could operate as a companion to Elastic Security rather than as a replacement. Elastic continued to collect, detect and store the events, while LogLookup AI focused on normalisation, correlation, risk scoring, evidence presentation and investigation support. Together, the lab shows a complete and understandable SOC workflow from raw endpoint activity to a structured incident.